CHAPTER 11

Random Variables and Probability Functions

Outline

11.1 Introduction

In Chapter 4, we described frequency distributions as a useful way of summarizing variations in observed data. Probability distributions are related to frequency distributions. In chapter 11, we developed techniques for calculating probability. If we can determine the probability of every possible outcome of an experiment, we can model the relative frequency distribution for the experiment. In chapter 11, we constructed sample spaces and determined probabilities of differer events in various random experiments. If a coin is tossed 2 or three times or a die is rolled 1 or times, it is simple to construct a sample space of such experiment. But if a coin is tossed 10 time then number of simple events or outcomes would be 2^{10} or 1024, or if a die is rolled 5 times, t number of outcomes would be 6^5 or 7776. Eventually the sample space of such experiment is ve difficult, time consuming and in most of the practical cases, is not usually possible to construct.

it is also difficult to compute probabilities of different events of such experiment. Such complicated tasks can easily be handled with random variables and their distributions. Random variables and their distributions (probability distributions) play a fundamental role in statistics. In this chapter, we present the concept of random variables and their probability distributions.

11.2 Random variable

Tossing a coin. Consider an experiment of tossing a single coin. The sample space of this experiment is $S=\{H,T\}$. Suppose we would like to know the number of heads in the observed outcome. Let X be a function that counts the number of heads in each outcome. Then $X(w)=1$ if $w=H$, and $X(w)=0$ if $w=T$. Here the values of X are real, that is, X is a real-valued function. We call X a random variable (or stochastic variable). Thus the possible values of random variable X are 0, 1. Upper case Latin letters such as X, Y, Z, etc. denote random variables and lower case letters x, y, z etc. denote the corresponding values of the random variables.

Definition 11.2.1 **A random variable** is a function that assigns a real number to each outcome in the sample space.

Example 11.1 Suppose a coin is tossed twice. The sample space,

$S = \{HH, HT, TH, TT\}$.

Let X denote the number of heads.

$$\begin{array}{lcccc} \text{Outcomes :} & HH & HT & TH & TT \\ & \downarrow & \searrow & \swarrow & \downarrow \\ x: & 2 & \multicolumn{2}{c}{1} & 0 \end{array}$$

That is, X is defined by

$X(w)=$ number of H's in w of S.

Then, $X(HH)=2$, $X(HT)=X(TH)=1$, and $X(TT)=0$.

Thus, X is a random variable which can assume values 0, 1 and 2.

Example 11.2. Suppose a point is randomly selected on the surface of a lake that has a maximum depth of 200 feet. Minimum depth may be 0 feet. Thus, sample space $S = [0, 200]$. Let Y denote the depth of the lake at the randomly chosen point. Then Y is a random variable and possible values of Y are the numbers in the interval [0, 200].

The set of all possible values of a random variable is called its **space**.

If we look on the random variable listed in Example 11.1, we observe that random variable, *number of heads*, takes the values 0, 1, 2 making certain gaps. It is mentioned that the possible values of the variable are countable numbers. The variable assumes a sequence of isolated or separated points on the number line shown by solid circle (See Figure 11.1). The random variable discussed here is discrete.

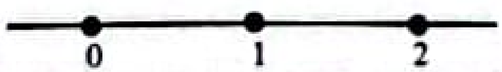


Figure 11.1 Number of heads: Discrete variable

Again, if we look on the variable listed in Example 11.2, we observe that random variable, depth, could take on for example, any value between 0 feet and 200 feet. It is mentioned that the possible values of the variable cannot be measured by count. The variable can take on any value on some interval (there are no breaks between possible values) (See Figure 11.2). The random variable discussed here is continuous.

Figure 11.2 Depth: Continuous variable

Thus, like quantitative variables discussed in Chapter 2, random variables can be classified as ***discrete*** and ***continuous***, according to the values those random variables can assume.

Definition 11.2.2 A **discrete random variable** is one that can assume a sequence of separated points on the real line. In other words, if the space of a random variable is countable, then the random variable is called a discrete random variable.

Example 11.3

a. Number of red balls on a randomly selected two balls without replacement from an urn that contains 2 red and 2 black balls is a discrete random variable.

The sample space of this experiment is

$$S = \{BB, BR, RB, RR\}.$$

Let X denote the number of red balls. Thus, X is defined as

$$X(w) = \text{Number of } R\text{'s in } w \text{ of } S$$

Then, $X(BB)=0$, $X(BR)=X(RB)=1$, and $X(RR)=2$.

Thus, X is a discrete random variable which can assume values 0, 1 and 2.

b. Number of telephone calls received by a telephone booth during a randomly selected time period is a discrete random variable.

The sample space of this experiment is

$$S = \{w \mid w = \text{number of telephone calls received}\}$$

Let Y denote the number of telephone calls received. Thus, Y is defined as

$Y(w)=0$ if $w = 0$ call received, $Y(w)=1$ if $w = 1$ calls received, $Y(w)=2$ if $w = 2$ calls received and so on.

Thus, Y is a discrete random variable which can assume values $0, 1, 2, \cdots$

Some examples of discrete random variables are:

- Number of defective bulbs on a randomly selected piece of bulbs from a production line
- Number of under-five children in a randomly selected family
- Number of credit hours taken by a randomly selected student this semester
- Number of draws until the last black ball is drawn when balls are drawn without replacement from a box containing 3 red and 2 black balls

Definition 11.2.3 A **continuous random variable** is one that can assume any value in an interval or a union of intervals on a line. In other words, a random variable is called continuous if its space is either an interval or a union of intervals.

Example 11.4

a. Consider an experiment about the demand for water in a certain city per day. The minimum demand for water is 4 million litters and maximum demand is 200 million litters. Then the sample space of this experiment is

$S = \{ w \mid w \text{ is the demand for water such that } 4 \leq w \leq 200 \}$.

Let X denote the demand for water of a randomly selected day. Thus, X is defined as

$X(w) = x$, when $w = x$.

The possible values of X are the numbers in the interval [4, 200]. In this example, X is a continuous random variable.

b. Consider an experiment that measure the waiting time in minutes at a supermarket checkout counter. Let Y denote the waiting time to check out. Then Y is a continuous random variable since the possible values of Y are any numbers in the interval $(0, \infty)$.

Some examples of discrete random variables are:

- Cumulative grade point average on a 0-4 point scale of a randomly selected student
- Demand for electricity in a certain city of a randomly selected day
- Length of a randomly selected light bulb (measured in hours) from a production line
- Body temperature of a randomly selected patient from a hospital

Example 11.5 Suppose we randomly select a student attending a university. Classify each of the following random variables as discrete or continuous:

a. Number of credit hours taken by the student this semester

b. Cumulative grade point of the student

Solution **a.** The number of credit hours taken by the student is a discrete random variable because it can assume only a countable number of values (for example, 15, 16, 17, and so on).

b. The cumulative grade point for the student is a continuous random variable since it could theoretically assume any value (for example 3.54678) corresponding to the points on the line interval from 0 to 4.

Theorem 1.2.1 Let X be a random variable. Then a real valued function $Y = g(X)$ is also a random variable.

11.3 Probability Distribution

Let S be a sample space of a random experiment and a probability measure has been specified on S. Let X be a random variable defined on S. We can determine probabilities associated with the possible values of X. Let C be a subset of the real line such that $\{X \in C\}$ is an event. Let $P(X \in C)$ denote the probability that the value of X will belong to C. Then $P(X \in C)$ is equal to the probability that the outcome $w \in S$ will be such that $X(w) \in C$. In symbols,

$P(X \in C) = P(\{w : X(w) \in C\})$.

Definition 11.3.1 The probability distribution of a random variable X is the collection of all probabilities of the form $P(X \in C)$ for all sets C of real numbers such that $X \in C$ is an event.

A probability distribution may be either discrete or continuous according as the corresponding random variable is discrete or continuous. These distributions will be discussed in the following sections.

11.4 Discrete Probability Distribution or Probability Function

In the tossing of two fair coins, let X denote the number of heads. Then X can assume values 0, 1, and 2. We can associate these values with probabilities in the following way:

$$P(X=0)=P(TT)=\frac{1}{4}$$

$$P(X=1)=P(HT,TH)=P(HT)+P(TH)=\frac{2}{4}$$

$$P(X=2)=P(HH)=\frac{1}{4}.$$

We can write this in the tabular form

Table 11.1

x	0	1	2
$f(x)=P(X=x)$	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{1}{4}$

The resulting distribution is a discrete probability distribution or probability function or probability mass function.

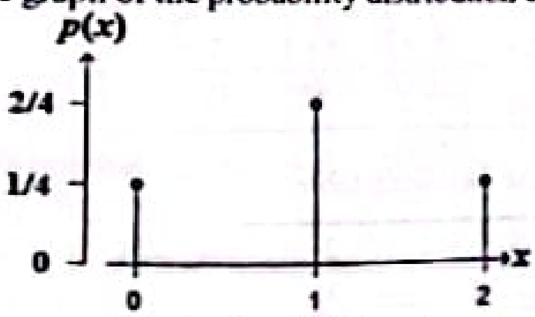
If we repeat the two-coin tossing experiment infinite number of times and record number of heads in each toss, most of the relative frequencies differ a bit from the probabilities of the above table. That is, the relative frequency distribution of X would appear as shown in **Table 11.1.**

Frequently, it is convenient to represent the probability distribution in the tabular formula as

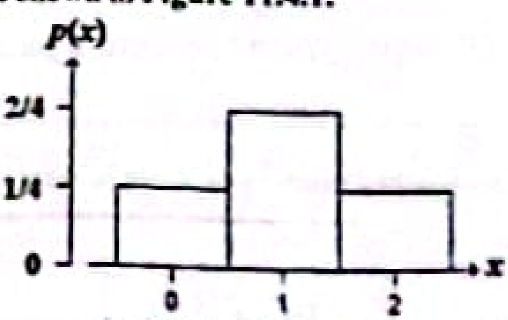
$$f(x)=P(X=x)=\begin{cases}\binom{2}{x}\left(\frac{1}{2}\right)^2, & \text{for } x=0,\ 1,\ 2,\\ 0, & \text{otherwise.}\end{cases}$$

It is clear that $f(x)\geq 0$, and $f(0)+f(1)+f(2)=\frac{1}{4}+\frac{2}{4}+\frac{1}{4}=1.$

The graph of the probability distribution can also be shown in **Figure 11.4.1.**



(a): Probability line graph



(b): Probability histogram

Figure 11.4.1

Definition 11.4.1. Let X be a random variable with range space R_X. Then the function f is defined by

$$f(x) = P(X = x)$$

for all $x \in R_X$

is called **discrete probability distribution** or probability function. If x is not one of the possible values of X, then $f(x) = 0$.

The values of a discrete random variable are often called mass points, and $f(x)$ denotes the mass associated with the mass point x. So, probability function is often called **probability mass function**.

A probability function $f(x)$ of a discrete random variable X satisfies the following two properties:

i) $f(x) \geq 0$ for all x, and

ii) $\sum_{\text{All } x} f(x) = 1$.

For each set C,

$$P(X \in C) = \sum_{x \in C} f(x).$$

Example 11.6 Drawing Batteries. Suppose that 2 batteries are randomly chosen from a bin containing 10 batteries, of which 7 are good and 3 are defective. Let X denote the number of defective batteries chosen. Find the probability function of X. Also determine

a. $P(X > 0)$, b. $P(X \leq 1)$, c. $P(X = 1)$.

Solution The random variable X takes the values 0, 1, or 2. Space of X is $R_X = \{0, 1, 2\}$

Now, $f(0) = P(X = 0) = P(0 \text{ defective, 2 good batteries}) = \dfrac{\binom{3}{0}\binom{7}{2}}{\binom{10}{2}} = \dfrac{1 \times 21}{45} = \dfrac{21}{45}$

$$f(1) = P(X = 1) = P(1 \text{ defective, 1 good batteries}) = \frac{\binom{3}{1}\binom{7}{1}}{\binom{10}{2}} = \frac{3 \times 7}{45} = \frac{21}{45}$$

$$f(2) = P(X = 2) = P(2 \text{ defective, 0 good batteries}) = \frac{\binom{3}{2}\binom{7}{0}}{\binom{10}{2}} = \frac{3 \times 1}{45} = \frac{3}{45}$$

Thus the probability function of X can be summarized in the following table:

x	0	1	2
$f(x) = P(X = x)$	$\frac{21}{45}$	$\frac{21}{45}$	$\frac{3}{45}$

The probability function of X can also be expressed as

$$f(x)=\begin{cases}\dfrac{\binom{3}{x}\binom{7}{2-x}}{\binom{10}{2}} & \text{for } x=0,1,2,\\ 0 & \text{otherwise.}\end{cases}$$

a. $P(X>0)=P(X=1)+P(X=2)=\frac{21}{45}+\frac{3}{45}=\frac{24}{45}.$

b. $P(X\le 1)=P(X=0)+P(X=1)=\frac{21}{45}+\frac{21}{45}=\frac{42}{45}.$

c. $P(X=1)=\frac{21}{45}.$

Example 11.7 Drawn Balls until last black ball. A box contains 2 black and 3 red balls. Balls are drawn successively without replacement until the last ball is black. If the random variable X denotes the number of draws until the last black is obtained, find the probability function of X. Also find **a.** $P(X=6)$, **b.** $P(X\ge 3)$, **c.** $P(2\le X\le 3)$.

Solution Let 'B' denote the balck ball and 'R' the white ball. The sample space of this experiment is given by

$$S=\{BB, BRB, RBB, BRRB, RBRB, RRBB, BRRRB, RBRRB, RRBRB, RRRBB\}$$

Hence the sample space has 10 points and all sample points are equally probable.

The random variable X takes the values 2, 3, 4, and 5. That is, space of X is $R_x=\{2,3,4,5\}$.

We can associate the values of X with probabilities in the following way:

$$f(2)=P(X=2)=P(\{BB\})=\frac{1}{10}$$

$$f(3)=P(X=3)=P(\{BRB, RBB\})=\frac{2}{10}$$

$$f(4)=P(X=4)=P(\{BRRB, RBRB, RRBB\})=\frac{3}{10}$$

$$f(5)=P(X=5)=P(\{BRRRB, RBRRB, RRBRB, RRRBB\})=\frac{4}{10}.$$

Thus the probability function of X can be summarized in the following table:

x	2	3	4	5
$f(x)=P(X=x)$	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{4}{10}$

The probability function of X can also be expressed as

$$f(x)=\begin{cases}\dfrac{x-1}{10} & \text{for } x=2,3,4,5,\\ 0 & \text{otherwise.}\end{cases}$$

a) $P(X=6)=P(\phi)=0$

b) $P(X\geq 3)=1-P(X\leq 2)=1-P(X=2)=1-\frac{1}{10}=\frac{9}{10}=0.90$

c) $P(2\leq X\leq 3)=P(X=2)+P(X=3)=\frac{1}{10}+\frac{2}{10}=\frac{3}{10}=0.30$

Example 11.8 The probability function of a discrete random variable X is

$$f(x)=\begin{cases} a\left(\frac{5}{6}\right)^x & \text{for } x=0,1,2,3,\cdots \\ 0 & \text{otherwise.}\end{cases}$$

Evaluate a and $P(X<3)$.

Solution Since $f(x)$ is a probability function, $\sum_{x=0}^{\infty} f(x)=1$.

Now, $f(0)=a\left(\frac{5}{6}\right)^0=a$, $f(1)=a\left(\frac{5}{6}\right)$, $f(2)=a\left(\frac{5}{6}\right)^2$, $f(3)=a\left(\frac{5}{6}\right)^3$, and so on.

Hence, $\sum_{x=0}^{\infty} f(x)=a+a\left(\frac{5}{6}\right)+a\left(\frac{5}{6}\right)^2+a\left(\frac{5}{6}\right)^3+\cdots$

$$=a\left[1+\left(\frac{5}{6}\right)+\left(\frac{5}{6}\right)^2+\left(\frac{5}{6}\right)^3+\cdots\right]$$

$$=a\left(1-\frac{5}{6}\right)^{-1}=6a.$$

Thus, $6a=1$

which gives, $a=\frac{1}{6}$.

Hence the complete probability function of X is

$$f(x)=\begin{cases} \frac{1}{6}\left(\frac{5}{6}\right)^x & \text{for } x=0,1,2,3,\cdots \\ 0 & \text{otherwise.}\end{cases}$$

Now, $P(X<3)=f(0)+f(1)+f(2)=\frac{1}{6}+\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)+\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^2=\frac{91}{216}$.

11.4.1 Graph of Discrete Probability Distribution

It is often helpful to represent a probability function in graphic form. The discrete probability distribution can be graphed in the form of a bar chart or a histogram. A graph of the probability distribution displays the possible values of a discrete random variable on the horizontal axis and the probabilities of those values on the vertical axis. In probability bar chart, the probability of each value is represented by a vertical line whose height equals the probability (See Figure 11.4.1(a) on page 479). In probability histogram the probability of each value is represented by a vertical rectangle, leaving space between rectangles, with height equals the corresponding probability. The bases of rectangles are of equal width and each base is centered with corresponding value of the random variable (See Figure 11.4.1(b) on page 479).

11.5 Continuous Probability Distribution or Probability Density Function

For a discrete random variable X, we can assign a positive probability to each value that X can take and get the probability distribution of X. The sum of all probabilities of the distribution is 1. However, not all experiments result in discrete random variables. Continuous random variables, for example, heights, weights, longevity of light bulb, demand of electricity, or experimental error of a laboratory, can assume infinitely many values. If we try to assign positive probability to each of the uncountable values, the probabilities will no longer sum to 1. Therefore, we must need a different approach to generate the probability distribution of a continuous random variable.

Suppose we have a data set on a continuous random variable, and we construct a relative frequency histogram to describe its distribution. For a small data set, we could use a small number of classes. A typical relative histogram is given for this small data set shown in Figure 11.5.1(a). As more and more observations are collected, we could use more classes reducing class width. The shape of histogram will change slightly, for the most part becoming less and less irregular, as shown in Figure 11.5.1(b). As the data set becomes very large and the class widths become very narrow, the relative frequency histogram appears more and more like the smooth curve shown in Figure 11.5.1(c). That is, the sequence of relative histograms approaches a smooth curve.

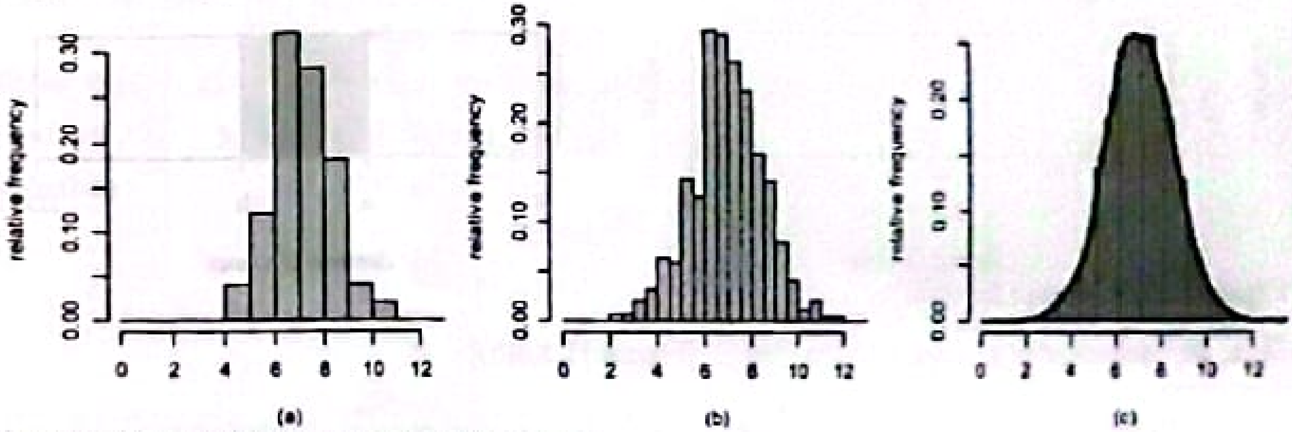


Figure 11.5.1 Relative frequency histogram

How can we create a model for this probability distribution? A continuous variable can take on any one of the infinite numbers of values on the real line. The probability distribution is created by one unit of probability along the line. A typical probability distribution is sketched in Figure 11.5.2.

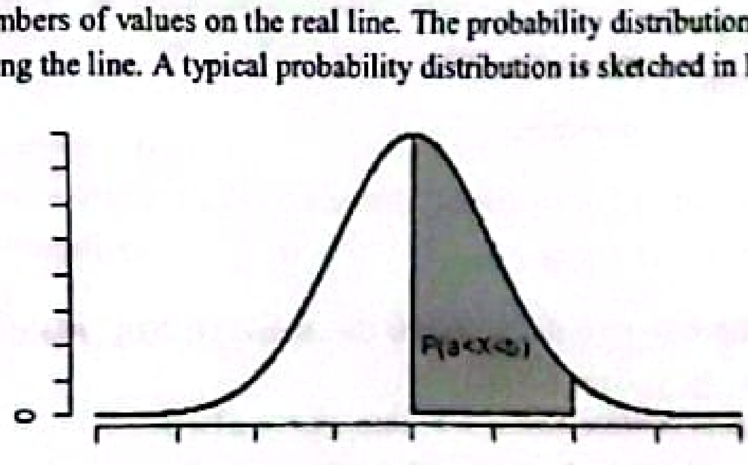


Figure 11.5.2 Density curve

The density of the probability that varies with values of X may be described by a mathematical formula $f(x)$. This $f(x)$ is called the probability distribution or probability density function of X. The curve cannot go below the horizontal measurement scale, and the total area under the curve is 1. The probability that X falls in an interval (a, b) such as $P(a < X < b)$ is the area under the curve equal to the shaded region.

In Example 11.4, We defined the random variable of the demand for water. Then the sample space of this experiment is

$$S = \{ w \mid w \text{ is the demand for water such that } 4 \le w \le 200 \}.$$

The possible values of the random variable X are the numbers in the interval [4, 200]. The distribution of X consists of all probabilities of the form $P(X \in C)$ for all sets C such that $X \in C$ is an event.

The distribution of demand for water is shown in Figure 11.5.3 (a).

In particular, C is a subinterval of the interval [4, 200]. Then

$$P(X \in C)\frac{1 \times \text{length of interval } C}{1 \times \text{length of interval } [4, 200]}$$

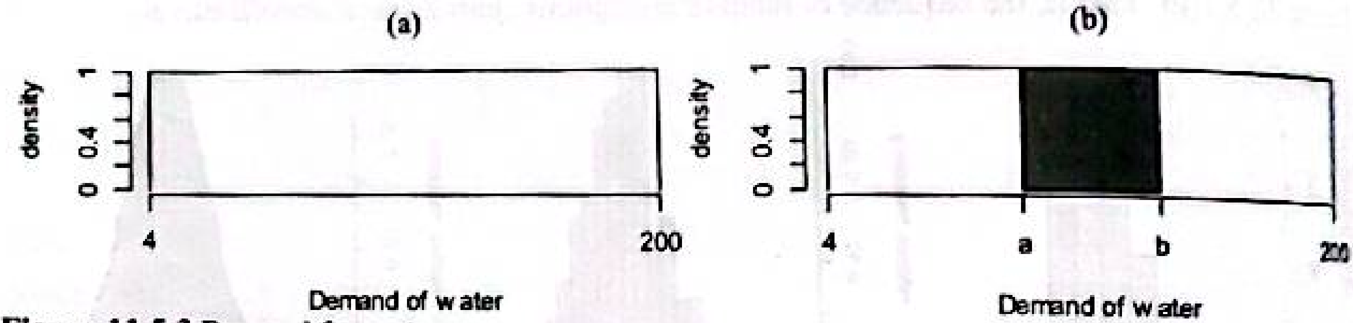


Figure 11.5.3 Demand for water

Thus, for each interval $C = [a, b] \subset [4, 200]$, [Figure 11.5.3(b)]

$$P(a \le X \le b) = \frac{b - a}{200 - 4} = \int_a^b \frac{1}{196} dx .$$

So, if we define

$$f(x) = \begin{cases} \frac{1}{196} & \text{for } 4 \le x \le 200, \\ 0 & \text{elsewhere,} \end{cases}$$

we then have

$$P(a \le X \le b) = \int_a^b f(x) dx .$$

Since $f(x)$ is defined to be 0 for x outside the interval [4, 200], $P(a \le X \le b)$ holds for all $a \le b$, even if $a = -\infty$ and (or) $b = \infty$.

For continuous random variable X, if $a = b$, then $P(X = a) = 0$.

Hence, $P(a \le X \le b) = P(a < X \le b) = P(a \le X < b) = P(a < X < b)$.

In fact, $P(X = a) = 0$ does not necessarily imply that $X = a$ is impossible.

Definition 11.5.1 If for a continuous random variable X, there exists a *non-negative function* f, defined on a real line, such that for every real interval C,

$$P(X \in C) = \int_C f(x)dx ,$$

then f is called the **density function (p.d.f)** of X.

A p.d.f. must satisfy the following two conditions:

i) $f(x) \geq 0$ for $-\infty < x < \infty$,

ii) $\int_{-\infty}^{\infty} f(x)dx = 1$.

The probability that X falls in any particular interval, say $[a, b]$, is the area under the density curve and above the interval. That is,

$$P(a \leq X \leq b) = \int_a^b f(x)dx .$$

Example 11.9 A random variable X has the following form

$$f(x) = \begin{cases} \dfrac{2}{x^2} & \text{for } 1 < x < 2, \\ 0 & \text{otherwise.} \end{cases}$$

Show that f is a density function for some random variable X.

Find $P(X < 1.5)$, $P(X > 1.5)$, and $P(0 < X \leq 1.5)$.

Solution

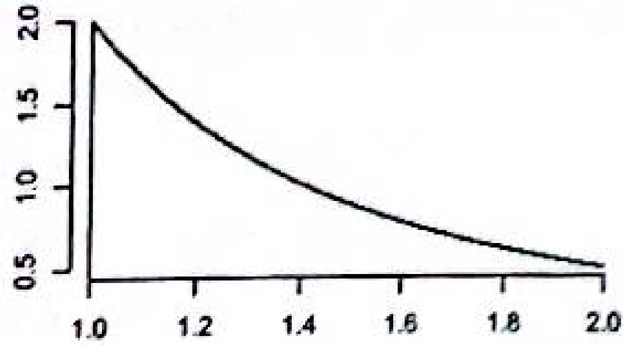


Figure 11.5.4 Density curve of $f(x)$

Since the domain or space of f is (1, 2), f is strictly positive on the interval and 0 outside this interval. Hence f is nonnegative.

Next, $\int_{-\infty}^{\infty} f(x)dx = \int_1^2 \frac{2}{x^2} dx$

$$= -2\left[\frac{1}{x}\right]_1^2$$

$$= -2\left[1-\frac{1}{2}\right]_1^2 = (-2)\left(-\frac{1}{2}\right) = 1.$$

Thus, f is the probability density function.

$$P(X < 1.5) = \int_{-\infty}^{1.5} f(x)dx = \int_1^{1.5} \frac{2}{x^2}dx = -2\left[\frac{1}{x}\right]_1^{1.5} = (-2)\left(\frac{1}{1.5}-1\right) = 0.667$$

$$P(X > 1.5) = 1 - P(X \le 1.5) = 1 - P(X < 1.5) = 1 - 0.667 = 0.333$$

$$P(0 < X \le 1.5) = \int_0^{1.5} f(x)dx = \int_1^{1.5} \frac{2}{x^2}dx = 0.667.$$

Example 11.10 A continuous random variable X has the function

$$f(x) = \begin{cases} k\dfrac{x^2}{\theta^3} & \text{for } 0 \le x \le \theta, \\ 0 & \text{elsewhere,} \end{cases}$$

for some $\theta > 0$.

a. For what value of k is f a p.d.f.?
b. What is the ratio of the mode to the median for this distribution? What is the shape of the distribution in context of skewness?
c. What is the probability of X less than the ratio of the mode to the median for this distribution?

Solution Note that $f(x) \ge 0$. Now, for $f(x)$ to be p.d.f, we need $\int_{-\infty}^{\infty} f(x)dx = 1$.

Now, $1 = \int_{-\infty}^{\infty} f(x)dx = \int_0^{\theta} \frac{kx^2}{\theta^3}dx$

$$= \frac{k}{\theta^3}\left[\frac{x^3}{3}\right]_0^{\theta}$$

$$= \frac{k}{\theta^3}\frac{\theta^3}{3} = \frac{k}{3}.$$

which implies

$k = 3$.

For fixed $\theta > 0$, the density function $f(x)$ is increasing. Thus, $f(x)$ is maximum at the right end point of the interval $[0, \theta]$. Hence the mode of this distribution is θ.

Now, we determine the median, M, of the distribution.

Then,

$$\frac{1}{2} = \int_{-\infty}^{M} f(x)dx = \int_0^{M} \frac{3x^2}{\theta^3}dx$$

$$= \frac{3}{\theta^3}\left[\frac{x^3}{3}\right]_0^{M} = \frac{M^3}{\theta^3}.$$

Thus, $M = \frac{\theta}{\sqrt[3]{2}}$.

Hence, the ratio of the mode to the median is

$$\frac{\text{mode}}{\text{median}} = \frac{\theta}{\theta / \sqrt[3]{2}} = \sqrt[3]{2}.$$

Since median is less than mode, the distribution is skewed to the left.

The probability that X is less than the ratio of the mode to the median of this distribution is

$$P\left(X < \sqrt[3]{2}\right) = \int_0^{\sqrt[3]{2}} \frac{3x^2}{\theta^3} dx$$

$$= \frac{3}{\theta^3}\left[\frac{x^3}{3}\right]_0^{\sqrt[3]{2}} = \frac{\left(\sqrt[3]{2}\right)^3}{\theta^3} = \frac{2}{\theta^3}.$$

11.6 Cumulative Distribution Function of Random Variables

Often we are required to know the probability that a random variable X takes on values less than or equal to any prescribed value x in the real line. That is, we require to find $P(X \le x)$. This probability is measured as a function of x and is called the cumulative distribution function or simply distribution function of X.

Demand for water. Consider again the demand for water X whose density is obtained as

$$f(x) = \begin{cases} \frac{1}{196} & \text{for } 4 \le x \le 200, \\ 0 & \text{otherwise.} \end{cases}$$

For any value of $x < 4$, $P(X \le x) = 0$, and for any value of $x > 200$, $P(X \le x) = 1$. Thus, an alternative function that is more directly related to probabilities associated with X is obtained as

$$F(x) = P(X \le x) = \int_{-\infty}^{x} f(t)dt$$

$$= \begin{cases} 0, & x < 4 \\ \int_4^x \frac{1}{196} dt & \text{for } 4 \le x \le 200, \\ 1, & \text{for } x > 200, \end{cases}$$

$$= \begin{cases} 0, & \text{for } x < 4, \\ \frac{1}{196}[t]_4^x & \text{for } 4 \le x \le 200, \\ 1 & \text{for } x > 200, \end{cases}$$

$$= \begin{cases} 0 & \text{for } x < 4, \\ \frac{x-4}{196} & \text{for } 4 \le x \le 200, \\ 1, & \text{for } x > 200. \end{cases}$$

So for example, $P(X \le 100) = \frac{100-4}{196} = \frac{94}{196} = 0.490$.

Definition 11.6.1 The **cumulative distribution function** or distribution function (c.d.f.) F of a random variable X is defined as a function

$$F(x) = P(X \le x) \quad \text{for } -\infty < x < \infty.$$

It should be emphasized that a cumulative distribution function is uniquely defined as above for every random variable X, regardless whether the distribution is discrete, or continuous, or mixed. If the c.d.f. is known, it can be used to find probabilities of events defined in terms of its corresponding random variable.

Properties of Distribution Function

1. For each value of x, $P(X > x) = 1 - F(x)$
2. For all real values of a and b, where $a < b$, $P(a < X \le b) = F(b) - F(a)$.
3. $\lim_{x\to-\infty} F(x) = F(-\infty) = 0$ and $\lim_{x\to\infty} F(x) = F(\infty) = 1$
4. $0 \le F(x) \le 1$ and $F(x)$ is a non-decreasing function of X. That is, if $a < b$, then $F(a) \le F(b)$.
5. For each value of x, $P(X < x) = F(x^-)$, where $\lim_{\substack{y\to x \\ y<x}} F(y) = F(x^-)$
6. For each value of x, $P(X = x) = F(x) - F(x^-)$
7. If X is a continuous random variable, then density function of X is $f(x) = \frac{d}{dx}F(x)$.

Proof of some properties

Proof 1: The events $\{X > x\}$ and $\{X \le x\}$ are disjoint and their union is the whole sample space S. Hence, $P(X > x) + P(X \le x) = 1$, which implies
$P(X > x) = 1 - P(X \le x) = 1 - F(x)$.

Proof 2: The events $a < X \le b$ and $X \le a$ are mutually exclusive,
$(a < X \le b) \cup (X \le a) = X \le b$
Therefore, $P(a < X \le b) + P(X \le a) = P(X \le b)$.
or, $P(a < X \le b) = P(X \le b) - P(X \le a) = F(b) - F(a)$.

Proof 3: Let the probability function of a discrete random variable X be
$f(x);\ x = x_1, x_2, \cdots, x_n$.
Without loss of generality, we assume that $x_1 < x_2 < \cdots < x_{n-1} < x_n$.
Since $-\infty < x_1$, the event $X \le -\infty$ is an impossible event which is ϕ.
Therefore, $F(-\infty) = P(X \le -\infty) = P(\phi) = 0$.
Again, since $x_n < \infty$, the event $X \le \infty$ is a sure event which is S.
Thus, $F(\infty) = P(X \le \infty) = P(S) = 1$

Proof 4: We know that $F(x) = P(X \le x)$ and probability of an event must lie between 0 and 1.
So, $0 \le P(X \le x) \le 1$.

Therefore, $0 \le F(x) \le 1$.

Again, since $a < b$, $F(b) - F(a) = P(a < X \le b) \ge 0$.

Therefore, $F(b) \ge F(a)$; that is $F(a) \le F(b)$.

Proof 7. Suppose X is a continuous random variable and its probability density function is $f(x)$. Then F is continuous at every x.

Now, $F(x) = \int_{-\infty}^{x} f(x)dx$. Therefore, $\frac{d}{dx}F(x) = f(x)$.

From the definition and properties of a c.d.f. $F(x)$ of discrete distribution, if $a < b$ and $P(a < X < b) = 0$, then $F(x)$ will be constant and horizontal over the interval $a < x < b$. Furthermore, at every point of x such that $P(X = x) > 0$, the c.d.f. will jump by the amount $P(X = x)$.

For density function, since the probability of each individual point x is zero, the c.d.f. will have no jumps. So, $F(x)$ is continuous over the entire real line.

Example 11.11 Consider again the experiment of tossing two fair coins. Let X denote the number of heads. The probability function of X is

$$f(x) = \begin{cases} 1/4 & \text{for } x = 0, \\ 1/2 & \text{for } x = 1, \\ 1/4 & \text{for } x = 2, \\ 0 & \text{elsewhere.} \end{cases}$$

Find the cumulative distribution function of X and plot its graph. Using distribution function, find $P(X = 2)$, $P(0 < X \le 2)$, $P(X < 2)$, and $P(X \le 1.5)$.

Solution Let F be the c.d.f. of X.

$$F(x) = P(X \le x), \quad -\infty < x < \infty.$$

It is easy to see that $F(x) = 0$ for $x < 0$.

For $0 \le x < 1$, $F(x) = P(X \le x) = P(X = 0) = f(0) = \frac{1}{4}$

For $1 \le x < 2$, $F(x) = P(X \le x) = P(X \le 1) = P(X = 0) + P(X = 1)$

$$= f(0) + f(1) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

For $x \ge 2$, $F(x) = P(X \le x) = P(X \le 2)$

$$= P(X = 0) + P(X = 1) + P(X = 2)$$

$$= f(0) + f(1) + f(2) = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1.$$

In summary,

$$F(x) = \begin{cases} 0 & \text{for } x < 0, \\ 1/4 & \text{for } 0 \le x < 1, \\ 3/4 & \text{for } 1 \le x < 2, \\ 1 & \text{for } x \ge 2. \end{cases}$$

The graph of the distribution function is shown in Figure 11.6.1.

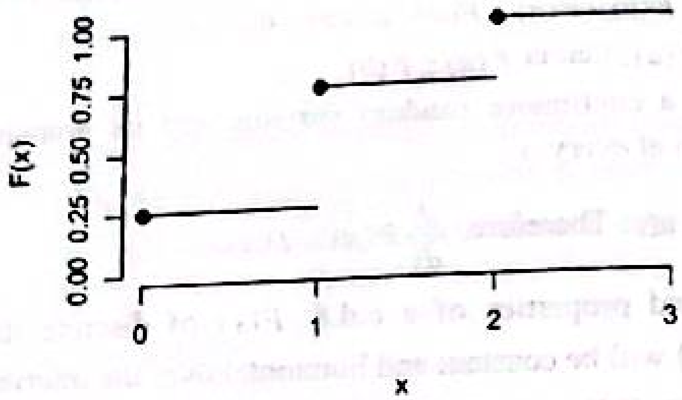


Figure 11.6.1 Distribution function

Using distribution function:

$$P(X = 2) = F(2) - F(2^-) = 1 - \frac{3}{4} = \frac{1}{4}.$$

$$P(0 < X \le 2) = F(2) - F(0^-) = 1 - 0 = 1.$$

$$P(X < 2) = F(2^-) = \frac{3}{4}.$$

$$P(X \le 1.5) = F(1.5) = \frac{3}{4}.$$

Example 11.12 Following is the cumulative distribution function of a random variable X.

$$\begin{cases} 0 & \text{for } x < -1, \\ 0.25 & \text{for } -1 \le x < 1, \\ 0.50 & \text{for } 1 \le x < 3, \\ 0.75 & \text{for } 3 \le x < 5, \\ 1 & \text{for } x \ge 5. \end{cases}$$

Graph the distribution function. Find the probability function of the random variable. Also find (a) $P(X \le 3)$, (b) $P(X = 3)$, (c) $P(X < 3)$, and (d) $P(2 \le X < 4)$.

Solution The graph of the distribution function is shown in Figure 11.6.2.

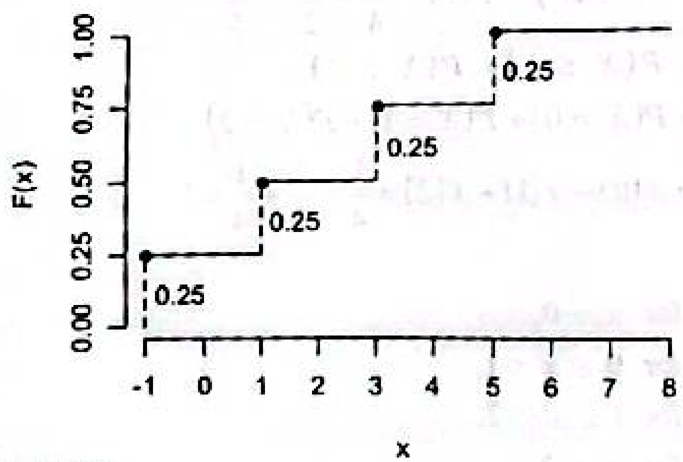


Figure 11.6.2 Distribution function

Let $f(x)$ be the probability function of the random variable X.

The space of the random variable is given by

$$R_X = \{-1, 1, 3, 5\}.$$

Now, $P(X = x) = F(x) - F(x^-)$.

$$f(-1) = P(X = -1) = F(-1) - F(-1^-) = 0.25 - 0 = 0.25$$

$$f(1) = P(X = 1) = F(1) - F(1^-) = 0.50 - 0.25 = 0.25$$

$$f(3) = P(X = 3) = F(3) - F(3^-) = 0.75 - 0.50 = 0.25$$

$$f(5) = P(X = 5) = F(5) - F(5^-) = 1 - 0.75 = 0.25.$$

Thus, the probability function of X is

$$f(x) = \begin{cases} 0.25 & \text{for } x = -1, \\ 0.25 & \text{for } x = 1, \\ 0.25 & \text{for } x = 3, \\ 0.25 & \text{for } x = 5, \\ 0 & \text{otherwise.} \end{cases}$$

(a) $P(X \le 3) = F(3) = 0.75$,

(b) $P(X = 3) = F(3) - F(3^-) = 0.75 - 0.50 = 0.25$,

(c) $P(X < 3) = F(3^-) = 0.50$,

(d) $P(2 \le X < 4) = F(4^-) - F(2^-) = 0.75 - 0.50 = 0.25$.

- The following things about the above distribution function, which are true in general, should be noted.

1. The magnitudes of the jumps at -1, 1, 3, 5 are 0.25, 0.25, 0.25, 0.25 which are precisely the probabilities. This fact enables one to obtain the probability function from the distribution function.
2. Because of the appearance of the graph of Figure 11.9, it is often called a staircase function or step function.
3. Unlike the c.d.f. of a discrete random variable, the c.d.f of a continuous random variable has no jump and is continuous everywhere.

Example 11.13 Suppose that X has the p.d.f.

$$f(x) = \begin{cases} 2x & \text{for } 0 < x < 1, \\ 0 & \text{elsewhere.} \end{cases}$$

Find and sketch the c.d.f. of X.

Find $P\left(X \le \frac{1}{4}\right)$, $P\left(X > \frac{1}{4}\right)$, $P\left(\frac{1}{3} < X \le \frac{2}{3}\right)$, $P\left(-1 \le X \le \frac{2}{3}\right)$, $P\left(X < \frac{2}{3}\right)$ and $P\left(X = \frac{2}{3}\right)$.

Solution Distribution function of random variable X is

$$F(x) = P(X \le x) = \int_{-\infty}^{x} f(t)\,dt$$

$$= \begin{cases} 0 & \text{for } x < 0, \\ \int_0^x 2t\,dt & \text{for } 0 \le x < 1, \\ 1 & \text{for } x \ge 1, \end{cases}$$

$$= \begin{cases} 0 & \text{for } x < 0, \\ \left[t^2\right]_0^x & \text{for } x \le 0 < 1, \\ 1 & \text{for } x \ge 1, \end{cases}$$

that is,

$$F(x) = \begin{cases} 0 & \text{for } x < 0, \\ x^2 & \text{for } x \le 0 < 1, \\ 1 & \text{for } x \ge 1. \end{cases}$$

The graph of the distribution function is shown in Figure 11.6.3.

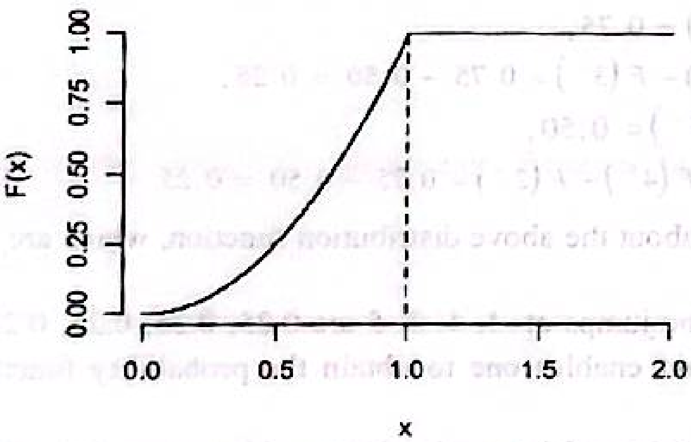


Figure 11.6.3 Distribution function

$$P\left(X \le \frac{1}{4}\right) = \int_0^{\frac{1}{4}} 2x\,dx = 2\left[\frac{x^2}{2}\right]_0^{\frac{1}{4}} = \left(\frac{1}{4}\right)^2 - (0)^2 = \frac{1}{16}.$$

Using distribution function, this probability can be calculated as follows:

$$P\left(X \le \frac{1}{4}\right) = F\left(\frac{1}{4}\right) = \left(\frac{1}{4}\right)^2 = \frac{1}{16}.$$

$$P\left(\frac{1}{3} < X \le \frac{2}{3}\right) = \int_{\frac{1}{3}}^{\frac{2}{3}} 2x\,dx = 2\left[\frac{x^2}{2}\right]_{\frac{1}{3}}^{\frac{2}{3}} = \left(\frac{2}{3}\right)^2 - \left(\frac{1}{3}\right)^2 = \frac{1}{3}.$$

Using distribution function,

$$P\left(\frac{1}{3} < X \le \frac{2}{3}\right) = F\left(\frac{2}{3}\right) - F\left(\frac{1}{3}\right) = \left(\frac{2}{3}\right)^2 - \left(\frac{1}{3}\right)^2 = \frac{1}{3}.$$

$$P\left(-1 \le X \le \frac{2}{3}\right) = F\left(\frac{2}{3}\right) - F(-1^-) = \left(\frac{2}{3}\right)^2 - 0 = \frac{4}{9}.$$

$$P\left(X < \frac{2}{3}\right) = F\left[\left(\frac{2}{3}\right)^-\right] = \left(\frac{2}{3}\right)^2 = \frac{4}{9}.$$

$$P\left(X = \frac{2}{3}\right) = F\left(\frac{2}{3}\right) - F\left[\left(\frac{2}{3}\right)^-\right] = 0.$$

Example 11.14 Let X denote the amount of gravel sold (in tons) during a randomly selected week at a particular sales facility. Suppose the density curve has height $f(x)$ above the value x, where

$$f(x) = \begin{cases} k(1-x) & \text{for } 0 \le x \le 1, \\ 0 & \text{elsewhere.} \end{cases}$$

a) Determine k for which $f(x)$ is a density function.

b) Find the distribution function of X and sketch its graph.

Using distribution function, evaluate the probability that

i) maximum amount of gravel sold is $1/2$ ton.

ii) amount of gravel sold is between $1/4$ ton and $1/2$ ton inclusive.

iii) minimum amount of gravel sold is $1/2$ ton.

Solution a) For $f(x)$ to be a density function, we have $\int_{-\infty}^{\infty} f(x) = 1$.

Thus

$$1 = k\int_0^1 (1-x)dx$$

$$= k\left[x - \frac{x^2}{2}\right]_0^1 = k\frac{1}{2}$$

from which

$$k = 2.$$

b) Distribution function of random variable X is

$$F(x) = P(X \le x) = \int_{-\infty}^{x} f(t)dt$$

$$= \begin{cases} 0 & \text{for } x < 0, \\ \int_0^x 2(1-t)dt & \text{for } 0 \le x < 1, \\ 1 & \text{for } x \ge 1, \end{cases}$$

$$= \begin{cases} 0 & \text{for } x < 0, \\ 2\left[t - \frac{t^2}{2}\right]_0^x & \text{for } 0 \le x < 1, \\ 1 & \text{for } x \ge 1, \end{cases}$$

that is,

$$F(x) = \begin{cases} 0 & \text{for } x < 0, \\ 2x - x^2 & \text{for } 0 \le x < 1, \\ 1 & \text{for } x \ge 1. \end{cases}$$

The graph of the distribution function is shown in Figure 11.6.4.

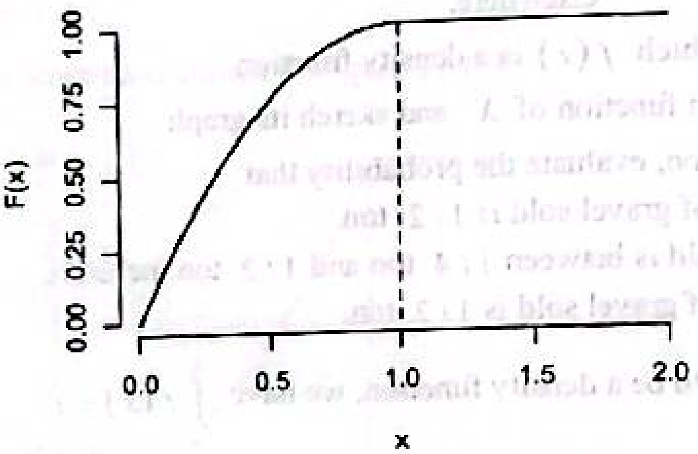


Figure 11.6.4 Distribution function

i) The probability that maximum amount of gravel sold is $1/2$ ton is

$$P\left(X \le \frac{1}{2}\right) = F\left(\frac{1}{2}\right) = 2\left(\frac{1}{2}\right) - \left(\frac{1}{2}\right)^2 = \frac{3}{4}.$$

ii) The probability that the amount of gravel sold is between $1/4$ ton and $1/2$ ton inclusive is

$$P\left(\frac{1}{4} \le X \le \frac{1}{2}\right) = F\left(\frac{1}{2}\right) - F\left(\frac{1^-}{4}\right)$$

$$= \left[2\left(\frac{1}{2}\right) - \left(\frac{1}{2}\right)^2\right] - \left[2\left(\frac{1}{4}\right) - \left(\frac{1}{4}\right)^2\right]$$

$$= \frac{3}{4} - \frac{7}{16} = \frac{5}{16}.$$

iii) The probability that the minimum amount of gravel sold is $1/2$ ton is

$$P\left(X \ge \frac{1}{2}\right) = 1 - P\left(X < \frac{1}{2}\right) = 1 - F\left(\frac{1^-}{2}\right)$$

$$= 1 - \left[2\left(\frac{1}{2}\right) - \left(\frac{1}{2}\right)^2\right] = 1 - \frac{3}{4} = \frac{1}{4}.$$

Example 11.15 A product has a warranty of one year. Let X be the time at which the product fails. The distribution function of the random variable X is as follows:

$$F(x) = \begin{cases} 0 & \text{for } x < 1, \\ 1 - \dfrac{1}{x^2} & \text{for } x \geq 1. \end{cases}$$

a) Find the density function of X.

b) What is the probability that the product would fail within two years?

c) What is the probability that the product would fail between 2 and 3 years?

Solution a) Since the value of the distribution is the same (0) at the point $x = 1$. Thus, the distribution is continuous. Hence, probability density function of X is

$$f(x) = \frac{d}{dx} F(x) = \begin{cases} \dfrac{d}{dx}\left(1 - \dfrac{1}{x^2}\right) & \text{for } x \geq 1, \\ 0 & \text{elsewhere,} \end{cases}$$

$$= \begin{cases} -2x^{-1}(-1) & \text{for } x \geq 1, \\ 0 & \text{elsewhere,} \end{cases}$$

$$= \begin{cases} \dfrac{2}{x} & \text{for } x \geq 1, \\ 0 & \text{elsewhere.} \end{cases}$$

b) The probability that the product would fail within two years is

$$P(X \leq 2) = F(2) = 1 - \left(\frac{1}{2}\right)^2 = \frac{3}{4}.$$

c) The probability that the product would fail between 2 and 3 years is

$$P(2 \leq X \leq 3) = F(3) - F(2^-)$$

$$= \left(1 - \frac{1}{3^2}\right) - \left(1 - \frac{1}{2^2}\right) = \frac{1}{4} - \frac{1}{9} = \frac{5}{36} = 0.139.$$

Example 11.16 The following is a function of a random variable X:

$$f(x) = \begin{cases} x & \text{for } 0 < x < 1, \\ 2 - x & \text{for } 1 \leq x < 2, \\ 0 & \text{elsewhere.} \end{cases}$$

a) Check whether it is a probability density function.

b) If the function is a density function, find the distribution function and sketch the graph.

c) Find $P(0.5 < x < 1.5)$ and $P(1 < x < 1.5)$.

Solution a) It is clear that $f(x) \geq 0$, and

$$\int_{-\infty}^{\infty} f(x)dx = \int_0^1 x\,dx + \int_1^2 (2 - x)\,dx$$

$$=\left[\frac{x^2}{2}\right]_0^1+\left[2x-\frac{x^2}{2}\right]_1^2=\frac{1}{2}+\left[(4-2)-\left(2-\frac{1}{2}\right)\right]=\frac{1}{2}+\frac{1}{2}=1.$$

Therefore, the given function is a probability density function.

b) By definition, the distribution function of X is

$$F(x)=P(X\le x)=\int_{-\infty}^{x}f(t)dt$$

For $x<0$,

$$F(x)=0.$$

For $0\le x<1$,

$$F(x)=\int_0^x t\,dt=\left[\frac{t^2}{2}\right]_0^x=\frac{x^2}{2}.$$

For $1\le x<2$,

$$F(x)=\int_0^1 t\,dt+\int_1^x(2-t)dt=\frac{1}{2}+\left[2t-\frac{t^2}{2}\right]_1^x=\frac{1}{2}+\left(2x-\frac{x^2}{2}\right)-\frac{3}{2}=2x-\frac{x^2}{2}-1.$$

For $x\ge 2$,

$$F(x)=1.$$

Thus,

$$F(x)=\begin{cases}0 & \text{for } x<0,\\ \dfrac{x^2}{2} & \text{for } 0\le x<1,\\ 2x-\dfrac{x^2}{2}-1 & \text{for } 1\le x<2,\\ 1 & \text{for } x\ge 2.\end{cases}$$

The graph of the distribution function is given in Figure 11.6.5.

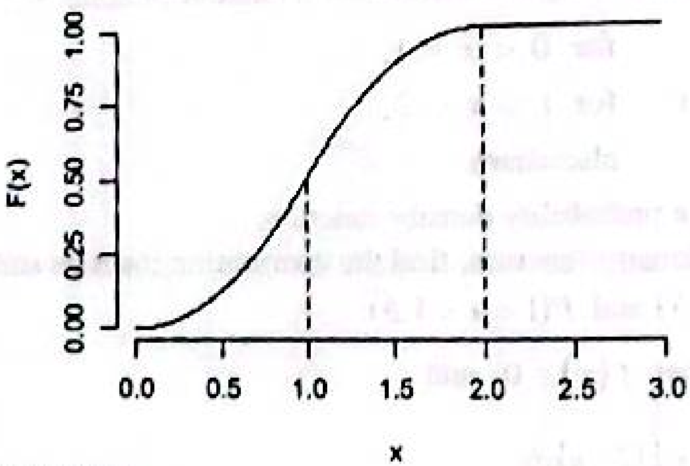


Figure 11.6.5 Distribution function

c) $P(0.5 < x < 1.5) = F(1.5^-) - F(0.5)$

$$= \left[2(1.5) - \left(\frac{1.5}{2}\right)^2 - 1\right] - (0.5)^2 = 0.875 - 0.125 = 0.75.$$

Example 11.17 Suppose the distribution function of a random variable X is given by

$$F(x) = \begin{cases} 0 & \text{for } x < 0, \\ \frac{x}{3} & \text{for } 0 \le x < 1, \\ \frac{1}{3} & \text{for } 1 \le x < 2, \\ \frac{x-1}{3} & \text{for } 2 \le x < 4, \\ 1 & \text{for } x \ge 4. \end{cases}$$

a. Is the distribution function continuous? If so, find the density function. If not, explain why not?
b. What is the probability that X is greater than 3?
c. What is the probability that X is between 1 and 3?
d. What is the probability that X is between 1 and 2?
e. What is the probability that X is greater than 3 given that X is greater than 2?

Solution

a. Since the probability of each individual point x is 0, the c.d.f. $F(x)$ has no jumps. Hence $F(x)$ is a continuous function over the entire real line. Hence the density function of X is continuous.
b. The probability that X is greater than 3 is

$$P(X > 3) = 1 - P(X \le 3) = 1 - F(3) = 1 - \frac{2}{3} = \frac{1}{3}.$$

c. The probability that X is between 1 and 3 is

$$P(1 < X < 3) = F(3^-) - F(1) = \frac{2}{3} - \frac{1}{3} = \frac{1}{3}.$$

d. The probability that X is between 1 and 2 is

$$P(1 < X < 2) = F(2^-) - F(1) = \frac{1}{3} - \frac{1}{3} = 0.$$

e. The probability that X is greater than 3 given that X is greater than 2 is

$$P(X > 3 \mid X > 2)$$

$$= \frac{P(X > 3 \cap X > 2)}{P(X > 2)}$$

$$= \frac{P(X > 3)}{P(X > 2)}$$

$$= \frac{1 - P(X \le 3)}{1 - P(X \le 2)} = \frac{1 - F(3)}{1 - F(2)} = \frac{1 - \frac{2}{3}}{1 - \frac{1}{3}} = \frac{1}{2}.$$

Example 11.18 Let the random variable X have the density function

$$f(x)=\begin{cases} kx & \text{for } 0<x<\sqrt{\dfrac{2}{k}}, \\ 0 & \text{elsewhere.} \end{cases}$$

If the mode of the distribution is at $x=\dfrac{\sqrt{2}}{4}$, then what is the median of X?

Solution The density function is an increasing function of X over its range or space. Thus, the density function is maximum at $x=\sqrt{\dfrac{2}{k}}$. Thus mode of the distribution is $\sqrt{\dfrac{2}{k}}$.

Hence, $\sqrt{\dfrac{2}{k}}=\dfrac{\sqrt{2}}{4}$

which implies $k=16$.

Let M be the median of the distribution.

Thus, $\dfrac{1}{2}=16\int_0^M x\,dx$

$$=16\left[\frac{x^2}{2}\right]_0^M=8M^2.$$

Therefore,

$$M^2=\frac{1}{16}$$

from which

$$M=\frac{1}{4}.$$

11.7 Two Dimensional Random Variable

In many random experiments, the outcomes involve more than one random variable. For example, a physician may study the joint behavior blood pressure and weight; an economist may study the joint behavior of business volume and profit. In fact, in most real life problems we come across will have more than one underlying random variable of interest. We confine our studies in the case of joint behavior of two random variables. Let us consider the following example.

Tossing a coin twice. Consider an experiment of tossing a single coin two times. The sample space of this experiment is $S=\{HH,HT,TH,TT\}$. Suppose we would like to know the number of heads and the number of tails in the observed outcome. Let X denote a function that counts the number of heads and Y denote another function that counts the number of tails in each outcome.

$$
\begin{array}{ccccc}
x & & \text{Outcomes} & & y \\
2 & \longleftarrow & HH & \longrightarrow & 0 \\
 & & HT & & \\
1 & \longleftarrow & & \longrightarrow & 1 \\
 & & TH & & \\
0 & \longleftarrow & TT & \longrightarrow & 2
\end{array}
$$

Thus X is defined as $X(w)=$ number of H's in w of S, and Y is defined as $X(w)=$ number of T's in w of S.

Then $X(w)=2$ and $Y(w)=0$ if $w=HH$, $X(w)=1$ and $Y(w)=1$ if $w=HT$ or $w=TH$, and $X(w)=0$ and $Y(w)=2$ if $w=TT$. When discussing two random variables at once, it is often convenient to put them together into an ordered pair, (X,Y). Here, the bivariate or two dimensional random variable (X,Y) can assume ordered pair of values $(2,0)$, $(1,1)$ and $(0,2)$.

Definition 11.7.1 A bivariate or **two dimensional random variable** is a function that assigns an ordered pair of real numbers to each outcome in the sample space.

We consider the random variables to be either both discrete or both continuous, or one discrete and the other continuous.

Definition 11.7.2 **A discrete bivariate random variable** (X,Y) is an ordered pair of two discrete random variables.

Example 11.19

a. A box contains 4 white balls, 3 black balls and 2 red balls. 2 balls are to be drawn without replacement. Let X denote the number of white balls and Y the number of black balls drawn. Then the ordered pair (X,Y) is a discrete bivariate random variable with $x=0,1,2$ and $y=0,1,2$, and $0 \le x+y \le 2$.

b. Suppose that in an electric display sign there are three light bulbs in the first row and four light bulbs in the second row. Let X denote the number of bulbs in the first row that will be burned out at a specified time, and Y the number of bulbs in the second row that will be burned out at the same time. Then the ordered pair (X,Y) is a discrete bivariate random variable with $x=0,1,2,3$ and $y=0,1,2,3,4$.

c. Suppose that a group of 8 executives of a certain firm include 6 who are married and 3 who never married. 3 of the executives are to be selected at random among which X denote the number of married executives and Y the number of never married executives. Then the ordered pair (X,Y) is a discrete bivariate random variable with $x=0,1,2,3$ and $y=0,1,2,3$.

Definition 11.7.3 **A continuous bivariate random variable** (X,Y) is an ordered pair of two continuous random variables.

Example 11.20

a. Consider an experiment about the demand for water and electricity per day. The water demand will be between 4 and 200 (in million of litters per day) and the electricity demand will be between 2 and 100 (million kilowatt-hours per day). Let X denote the water demand and Y the electricity demand at a random day. Then the ordered pair (X,Y) is a continuous bivariate random variable with $4 \le x \le 200$ and $2 \le y \le 100$.

b. Suppose a point is to be chosen at random from a square in the *xy*-plane with 1 cm length of each side. Then the ordered pair of point (X,Y) is a continuous bivariate random variable with $0 \le x \le 1$ and $0 \le y \le 1$.

c. A part of an electronic system has two types of components in joint operation. Let X and Y denote the random length of life (measured in hundreds of hours) of component of type I and of type II, respectively. Then the ordered pair (X,Y) is a continuous bivariate random variable with $x > 0$ and $y > 0$.

Definition 11.7.4 A mixed or hybrid bivariate random variable (X,Y) is an ordered pair of one discrete ,say X , and one continuous, say Y , random variables.

Example 11.21 Suppose in a clinical trial each patient with depression receives a treatment and is followed to see whether they have a relapse into depression. Let X be an indicator variable such that $X = 1$, if the first patient does not relapse and $X = 0$ if the patient relapses. Let Y be another variable that indicates the proportion of patients who have no relapse among all patients who might receive the treatment. Then the ordered pair (X,Y) is a mixed bivariate random variable with $x = 0,1$ and $0 \le y \le 1$.

11.8 Bivariate Distributions

We introduce the joint probability function for bivariate discrete random variable, the joint probability density function for bivariate continuous random variable. We also introduce a hybrid bivariate probability function for bivariate random variable (X,Y), where X is discrete and Y is continuous.

Definition 11.8.1 The **joint probability distribution** of a bivariate random variable (X,Y) is the collection of all probabilities of the form $P[(X,Y) \in C]$ for all sets C of pairs of real numbers such that $\{(X,Y) \in C\}$ is an event.

11.8.1 Discrete Joint Distributions

In section 11.6, the sample space S in an experiment of tossing a fair coin twice is given. X denotes the number of heads and Y the number of tails. Then (X,Y) can assume values $(2,0)$, $(1,1)$ and $(0,2)$. The simple events are equally likely. We can associate these values with probabilities in the following way:

$$P(X = 2, Y = 0) = P(HH) = \frac{1}{4}$$

$$P(X=1,Y=1)=P(HT,TH)=P(HT)+P(TH)=\frac{2}{4}$$

$$P(X=0,Y=2)=P(TT)=\frac{1}{4}.$$

We can write this in the tabular form (Table 11.2)

Table 11.2

(x,y)	$(2,0)$	$(1,1)$	$(0,2)$
$f(x,y)=P(X=x,Y=y)$	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{1}{4}$

The joint probability function can also been represented as Table 11.3.

Table 11.3

$x\downarrow$ \ $y\rightarrow$	0	1	2
0	0	0	$\frac{1}{4}$
1	0	$\frac{2}{4}$	0
2	$\frac{1}{4}$	0	0

Frequently, it is convenient to represent the joint probability function by a formula

$$f(x,y)=P(X=x,Y=y)=\begin{cases}\frac{1}{4}\binom{x+y}{x} & \text{for } x=0,1,2,\text{and } y=2-x,\\ 0 & \text{elsewhere.}\end{cases}$$

Definition 11.8.2 Let (X,Y) be a two dimensional discrete random variable with range spaces R_X and R_Y of X and Y, respectively. Then the function f defined by

$$f(x,y)=P(X=x,Y=y) \text{ for all } (x,y)\in R_X\times R_Y$$

is called **discrete joint probability function** or discrete bivariate probability function. If (x,y) is not one of the possible values of (X,Y), then $f(x,y)=0$.

A discrete joint probability function $f(x,y)$ satisfies the following two properties:

i) $f(x,y)\geq 0$ for all (x,y), and

ii) $\sum_{\text{All }(x,y)} f(x,y)=1$.

For each set C of ordered pairs,

$$P[(X,Y)\in C]=\sum_{(x,y)\in C} f(x,y).$$

Example 11.22 A box contains 4 white balls, 3 black balls and 2 red balls. 2 balls are to be drawn without replacement. Let X denote the number of white balls and Y the number of black balls drawn. Find the joint probability function of (X,Y). Find the probability that $X+Y \geq 3$. Also find the probability that $X=1$.

Solution In this case, X takes only the values 0, 1, and 2; and Y takes only the values 0, 1, and 2. The possible pair of values are (0, 0), (0, 1), (0, 2), (1,0), (1, 1), (2, 0).

$$f(0,0)=P(X=0,Y=0)=\frac{\binom{4}{0}\binom{3}{0}\binom{2}{2}}{\binom{9}{2}}=\frac{1}{36}.$$

$$f(0,1)=P(X=0,Y=1)=\frac{\binom{4}{0}\binom{3}{1}\binom{2}{1}}{36}=\frac{6}{36}.$$

$$f(0,2)=P(X=0,Y=2)=\frac{\binom{4}{0}\binom{3}{2}\binom{2}{0}}{36}=\frac{3}{36}.$$

$$f(1,0)=P(X=1,Y=0)=\frac{\binom{4}{1}\binom{3}{0}\binom{2}{1}}{36}=\frac{8}{36}.$$

$$f(1,1)=P(X=1,Y=1)=\frac{\binom{4}{1}\binom{3}{1}\binom{2}{0}}{36}=\frac{12}{36}.$$

$$f(2,0)=P(X=2,Y=0)=\frac{\binom{4}{2}\binom{3}{0}\binom{2}{0}}{36}=\frac{6}{36}.$$

We obtain the joint probability function of (X,Y) shown in the Table 11.4.

Table 11.4 Joint probability function

$y\rightarrow$ / $x\downarrow$	0	1	2
0	$\frac{1}{36}$	$\frac{6}{36}$	$\frac{3}{36}$
1	$\frac{8}{36}$	$\frac{12}{36}$	0
2	$\frac{6}{36}$	0	0

The joint discrete probability function can also be represent by a formula

$$f(x,y)=P(X=x,Y=y)=\begin{cases}\dfrac{\binom{4}{x}\binom{3}{y}\binom{2}{2-x-y}}{\binom{9}{2}} & \text{for } x=0,1,2; y=0,1,2; 0\le x+y\le 2,\\ 0 & \text{elsewhere.}\end{cases}$$

$P(X+Y\ge 3)$
$= f(1,2)+f(2,1)+f(2,2)=0+0+0=0.$
$P(X=1)$
$=\sum_{y=0}^{2} f(1,y) = f(1,0)+f(1,1)+f(1,2)$
$=\frac{8}{36}+\frac{12}{36}+0=\frac{20}{36}.$

The above joint discrete probability function is given in Figure 11.8.1.

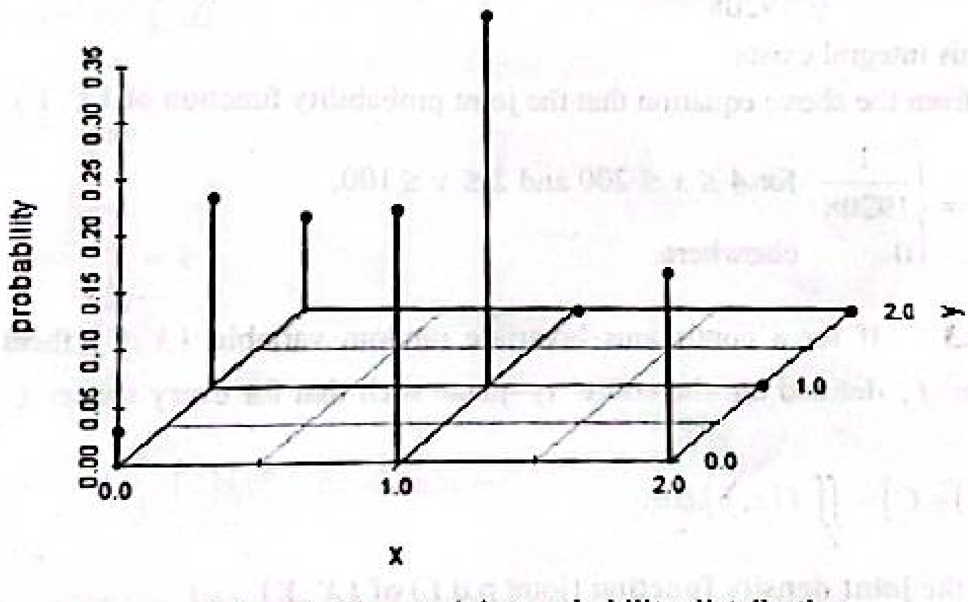


Figure 11.8.1 Probability graph for the discrete joint probability distribution

11.8.2 Continuous Joint Distributions

Consider an experiment about the demand for water and electricity per day. The water demand will be between 4 and 200 (in million of litters per day) and the electricity demand will be between 2 and 100 (million kilowatt-hours per day). The shaded region shows in Figure 11.8.2 shows the sample space of the experiment. Let X denote the water demand and Y the electricity demand at a random day. Then the ordered pair (X,Y) is a continuous bivariate random variable with $4\le x\le 200$ and $2\le y\le 100$.

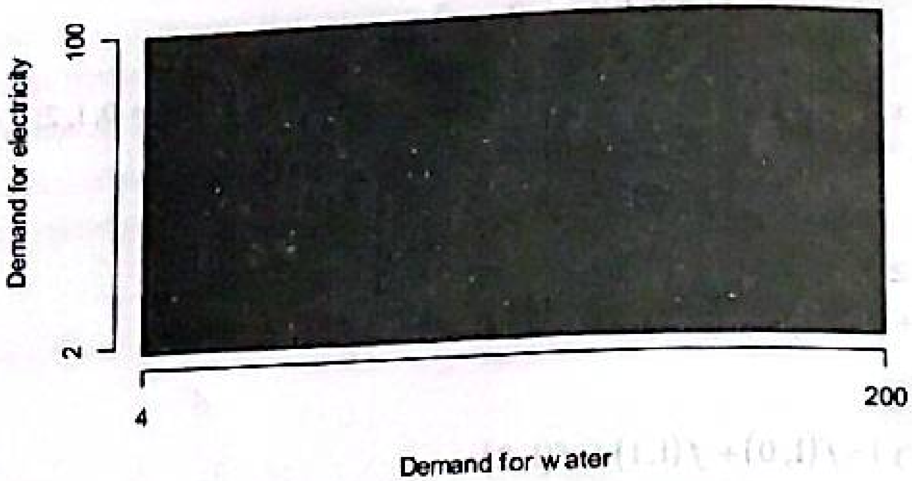


Figure 11.8.2 Joint density of demand for water and demand for electricity

We assumed that the probability of each subset of the sample space was proportional to the area of the subset. The area of the sample space is (200-4)(100-2) = 19208. The probability that the pair (X,Y) lies in a region C is the area of C divided by 19208. This relation can be expressed as

$$P[(X,Y)\in C]=\iint_C \frac{1}{19208}\,dxdy\,,$$

assuming that this integral exists.

Thus it is clear from the above equation that the joint probability function of (X,Y) is the function

$$f(x,y)=\begin{cases}\dfrac{1}{19208} & \text{for } 4\le x\le 200 \text{ and } 2\le y\le 100,\\ 0 & \text{elsewhere.}\end{cases}$$

Definition 11.8.3 If for a continuous bivariate random variable (X,Y), there exists an non-negative function f, defined on the entire xy-plane such that for every subset C of pairs of real numbers,

$$P[(X,Y)\in C]=\iint_C f(x,y)dxdy\,,$$

then f is called the **joint density function** (joint p.d.f.) of (X,Y).

A joint p.d.f. must satisfy the following two conditions:

i) $f(x,y)\ge 0$ for $-\infty<x<\infty$ and $-\infty<y<\infty$,

ii) $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dxdy=1$.

The probability that (X,Y) falls in any particular region C, say $[(a,b),(c,d)]$, of the plane is

$$P[(X,Y)\in C]=P(a\le X\le b, c\le Y\le d)=\int_a^b\int_c^d f(x,y)\,dxdy\,.$$

Example 11.23 Let the joint p.d.f. of the continuous bivariate random variable (X, Y) be

$$f(x,y)=\begin{cases} k\left(x^2+2xy\right) & \text{for } 0\le x\le 1,\ 0\le y\le 1,\\ 0 & \text{elsewhere.}\end{cases}$$

a) Determine the value of the constant k,

b) Find $P\left(X\le \frac{1}{2}, Y\ge \frac{1}{2}\right)$,

c) What is the probability of the event $(X\le Y)$?

Solution a) Since $f(x,y)$ is a density function,

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy=1.$$

It follows that

$$1=\int_0^1\int_0^1 k\left(x^2+2xy\right)dx\,dy$$

$$=k\int_0^1\left[\int_0^1\left(x^2+2xy\right)dy\right]dx$$

$$=k\int_0^1\left[x^2y+xy^2\right]_0^1 dx$$

$$=k\int_0^1\left[x^2+x\right]dx$$

$$=k\left[\frac{x^3}{3}+\frac{x^2}{2}\right]_0^1=k\frac{5}{6}.$$

Therefore, $k=\dfrac{6}{5}$.

b) $$P\left(X\le \frac{1}{2}, Y\ge \frac{1}{2}\right)=\frac{6}{5}\int_0^{1/2}\int_{1/2}^{1}\left(x^2+2xy\right)dx\,dy$$

$$=\frac{6}{5}\int_0^{1/2}\left[\int_{1/2}^{1}\left(x^2+2xy\right)dy\right]dx$$

$$=\frac{6}{5}\int_0^{1/2}\left[x^2y+xy^2\right]_{1/2}^{1}dx$$

$$=\frac{6}{5}\int_0^{1/2}\left[x^2(1-1/2)+x(1-1/4)\right]dx$$

$$=\frac{6}{5}\int_0^{1/2}\left(\frac{x^2}{2}+\frac{3x}{4}\right)dx$$

$$=\frac{6}{5}\left[\frac{x^3}{6}+\frac{3x^2}{8}\right]_0^{1/2}=\left(\frac{6}{5}\right)\left(\frac{1}{48}+\frac{3}{32}\right)=\frac{11}{80}.$$

c) $P(X \le Y)=\frac{6}{5}\int_0^1\left[\int_0^y\left(x^2+2xy\right)dx\right]dy$

$$=\frac{6}{5}\int_0^1\left[\frac{x^3}{3}+x^2y\right]_{x=0}^{x=y}dy$$

$$=\frac{6}{5}\int_0^1\left(\frac{y^3}{3}+y^3\right)dy$$

$$=\left(\frac{6}{5}\right)\left(\frac{4}{3}\right)\left[\frac{y^4}{4}\right]_0^1=\left(\frac{24}{15}\right)\left(\frac{1}{4}\right)=\frac{2}{5}.$$

11.9 Bivariate Cumulative Distribution Functions

The joint distribution function or joint cumulative distribution function (joint c.d.f.) F of a two dimensional random variable (X, Y) is defined as a function

$$F(x, y)=P(X \le x \text{ and } Y \le y),\ -\infty < x < \infty \text{ and } -\infty < y < \infty.$$

$F(x, y)$ is monotone increasing in x for each fixed y and is monotone increasing in y for each fixed x.

For given numbers $a < b$ and $c < d$,

$$\begin{aligned}
&P(a < X \le b \text{ and } c < Y \le d)\\
&= P(a < X \le b \text{ and } Y \le d)-P(a < X \le b \text{ and } Y \le c)\\
&= P(X \le b \text{ and } Y \le d)-P(X \le a \text{ and } Y \le d)\\
&- P(X \le b \text{ and } Y \le c)+P(X \le a \text{ and } Y \le c)\\
&= F(b, d)-F(a, d)-F(b, c)+F(a, c).
\end{aligned}$$

It is noted here that the c.d.f. of X is $F_1(x)=\lim_{y\to\infty} F(x, y)$. The c.d.f. of Y is $F_2(y)=\lim_{x\to\infty} F(x, y)$.

If the bivariate random variable (X, Y) is discrete, then

$$F(x, y)=P(X \le x \text{ and } Y \le y)=\sum_{r\le x}\sum_{s\le y} F(r, s) \text{ for } -\infty < x < \infty \text{ and } -\infty < y < \infty.$$

If the bivariate random variable (X, Y) is continuous, then

$$F(x, y)=P(X \le x \text{ and } Y \le y)=\int_{-\infty}^{y}\int_{-\infty}^{x} f(r, s)\,dr\,ds \text{ for } -\infty < x < \infty \text{ and } -\infty < y < \infty.$$

If the bivariate random variable (X, Y) is continuous, then

$$f(x, y)=\frac{\partial^2 F(x, y)}{\partial x \partial y}.$$

Example 11.24 With reference to Example 11.22, determine $F(1,1)$.

Solution $F(1,1) = P(X \le 1, Y \le 1)$

$$= f(0,0) + f(0,1) + f(1,0) + f(1,1)$$

$$= \frac{1}{36} + \frac{6}{36} + \frac{8}{36} + \frac{12}{36} = \frac{27}{36}.$$

Example 11.25 The joint p.d.f. of (X, Y) is

$$f(x,y) = \begin{cases} \frac{1}{8}(x+y) & \text{for } 0 < x < 2 \text{ and } 0 < y < 2, \\ 0 & \text{otherwise.} \end{cases}$$

The joint c.d.f. of (X, Y) is as follows:

For $x < 0$ or $y < 0$, $F(x,y) = 0$.

For $0 \le x < 2$ or $0 \le y < 2$,

$$F(x,y) = \int_0^x \int_0^y \frac{1}{8}(x+y)\,dy\,dx$$

$$= \int_0^x \frac{1}{8}\left(xy + \frac{y^2}{2}\right)dx = \frac{1}{16}xy(x+y).$$

For $0 \le x < 2$ and $y \ge 2$,

$$F(x,y) = \int_0^x \int_0^2 \frac{1}{8}(x+y)\,dy\,dx$$

$$= \int_0^x \frac{1}{4}(x+1)\,dx = \frac{1}{8}x(x+2).$$

For $x \ge 2$ and $0 \le y < 2$,

$$F(x,y) = \int_0^2 \int_0^y \frac{1}{8}(x+y)\,dy\,dx$$

$$= \int_0^2 \frac{1}{8}\left(xy + \frac{y^2}{2}\right)dx = \frac{1}{8}y(y+2)$$

For $x \ge 2$ and $y \ge 2$,

$$F(x,y) = 1.$$

Thus,

$$F(x,y) = \begin{cases} 0 & \text{for } x < 0 \text{ or } y < 0, \\ \frac{1}{16}xy(x+y) & \text{for } 0 \le x < 2 \text{ and } 0 \le y < 2, \\ \frac{1}{8}x(x+2) & \text{for } 0 \le x < 2 \text{ and } y \ge 2, \\ \frac{1}{8}y(y+2) & \text{for } x \ge 2 \text{ and } 0 \le y < 2, \\ 1 & \text{for } x \ge 2 \text{ and } y \ge 2. \end{cases}$$

Example 11.26 The joint c.d.f. for water and electric demand (X, Y) is

$$F(x,y)=\begin{cases}0 & \text{for } x<4 \text{ or } y<2,\\ \dfrac{(x-4)(y-2)}{19208} & \text{for } 4\le x<200 \text{ and } 2\le y<100,\\ \dfrac{x-4}{196} & \text{for } 4\le x<200 \text{ and } y\ge 100,\\ \dfrac{y-2}{98} & \text{for } x\ge 200 \text{ and } 2\le y<100,\\ 1 & \text{for } x\ge 200 \text{ and } y\ge 100.\end{cases}$$

a. Find the joint p.d.f. of (X, Y);

b. $P(50 \le X \le 100, 50 \le Y \le 75)$;

c. the c.d.f. of X,

d. the c.d.f. of Y.

Solution a. For $4 \le x \le 200$ and $2 \le y \le 100$,

$$f(x,y)=\frac{\partial^2 F(x,y)}{\partial x \partial y}=\frac{\partial^2}{\partial x \partial y}\left(\frac{(x-4)(y-2)}{19208}\right)$$

$$=\frac{\partial}{\partial x}\left(\frac{x-4}{19208}\right)=\frac{1}{19208}.$$

Thus,

$$f(x,y)=\begin{cases}\dfrac{1}{19208} & \text{for } 4\le x\le 200 \text{ and } 2\le y\le 100,\\ 0 & \text{elsewhere.}\end{cases}$$

a. $P(50 \le X \le 100, 50 \le Y \le 75)$

$$= F(100,75)-F(100,50)-F(50,75)+F(50,50)$$

$$=\frac{(100)(75)}{19208}-\frac{(100)(50)}{19208}-\frac{(50)(75)}{19208}+\frac{(50)(50)}{19208}=\frac{625}{9604}.$$

b. If $4 \le x \le 200$ and $y > 100$, then $F(x, y) = F(x, 100)$, and it follows that

$$F(x,y)=\frac{x-4}{196}.$$

Thus, by letting $y \to \infty$, the c.d.f. of X is

$$F_1(x)=\begin{cases}0 & \text{for } x<4,\\ \dfrac{x-4}{196} & \text{for } 4\le x<200,\\ 1 & \text{for } x\ge 200.\end{cases}$$

d) If $2 \le y < 100$ and $x > 200$, then $F(x, y) = F(x, 200)$, and it follows that

$$F(x,y) = \frac{y-2}{98}.$$

Thus, by letting $x \to \infty$, the c.d.f. of Y is

$$F_2(y) = \begin{cases} 0 & \text{for } y < 2, \\ \dfrac{y-2}{98} & \text{for } 2 \le y < 100, \\ 1 & \text{for } y \ge 100. \end{cases}$$

11.10 Marginal Distributions

In many occasions we know the joint c.d.f. or joint p.f./p.d.f. of a bivariate random variable (X, Y). We want to get the c.d.f. or p.f./p.d.f. of each of the variables in isolation. Such a process is called marginalization.

If the joint c.d.f. $F(x, y)$ of a bivariate random variable (X, Y) is known, then the c.d.f. $F_1(x)$ of random variable X can be derived from $F(x, y)$. An example of this derivation is given in Example 11.26 (c). If X has a continuous random variable, then we can also derive the p.d.f. of X from joint distribution.

If we observe carefully the middle two branches of the $F(x, y)$ in Example 11.26, we identified as $F_1(x)$ and $F_2(y)$, the c.d.f.'s of the two individual random variables X and Y. Now from the property of c.d.f., we have the p.d.f. of X as

$$f_1(x) = \frac{\partial F_1(x)}{\partial x} = \begin{cases} \dfrac{1}{196} & \text{for } 4 \le x \le 200, \\ 0 & \text{elsewhere.} \end{cases}$$

It is clear that the above $f_1(x)$ matches what we found in section 11.5. Similarly, the p.d.f. of Y is

$$f_2(y) = \frac{\partial F_2(y)}{\partial y} = \begin{cases} \dfrac{1}{98} & \text{for } 2 \le y \le 100, \\ 0 & \text{elsewhere.} \end{cases}$$

The above idea leads the following definition.

Definition 11.10.1 Let $F(x, y)$ be the joint c.d.f. of (X, Y). Then the **marginal c.d.f. of** X is $F_1(x) = \lim_{y \to \infty} F(x, y)$. Similarly, the **marginal c.d.f. of** Y is $F_2(y) = \lim_{x \to \infty} F(x, y)$. The p.f. or p.d.f. of X associated with the marginal c.d.f. of X is called the marginal p.f. or marginal p.d.f. of X.

The marginal p.f. and marginal p.d.f. can be obtained from joint p.f. and joint p.d.f., respectively.

The following diagram illustrates the concept of marginal graphically.

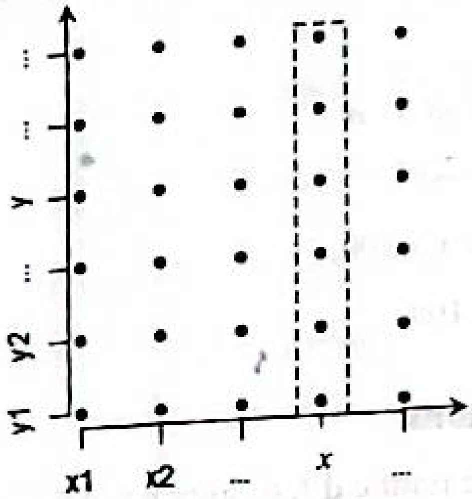


Figure 11.10.1 Computing $g(x)$ from Joint p.f.

In Figure 11.10.1, the set of point in the dashed box is the set of pairs with first coordinate x. The event $\{X = x\}$ can be expressed as the union of the disjoint events represented by the pairs in the dashed box, say, $E_y = \{X = x, \text{ and } Y = y\}$ for all possible y. That is, $\{X = x\} = \bigcup_{\text{All } y} E_y$.

Thus, $P(X = x) = \sum_{\text{All } y} P(E_y) = \sum_{\text{All } y} f(x, y)$.

Definition 11.10.2 Let $f(x, y)$ be a joint probability function of a discrete bivariate random variable (X, Y). Then the **marginal probability function** of X is

$$f_1(x) \text{ or } g(x) = \sum_{\text{All } y} f(x, y), \text{ for all } x.$$

Similarly, the marginal probability function of Y is

$$f_2(y) \text{ or } h(y) = \sum_{\text{All } x} f(x, y) \text{ for all } y.$$

Example 11.27 With reference to the joint probability function of Example 11.22, determine the marginal probability function of X and Y, where X denotes the number of white balls and Y the number of black balls drawn.

Solution The joint probability function of (X, Y) is shown in Table 11.5. Row and column totals are added to the margin.

Table 11.5 Joint and marginal probability functions

	y			
x	0	1	2	Row total
0	$\frac{1}{36}$	$\frac{6}{36}$	$\frac{3}{36}$	$\frac{10}{36}$
1	$\frac{8}{36}$	$\frac{12}{36}$	0	$\frac{20}{36}$
2	$\frac{6}{36}$	0	0	$\frac{6}{36}$
Column total	$\frac{15}{36}$	$\frac{18}{36}$	$\frac{3}{36}$	1

The marginal p.f. $f_1(x)$ or $g(x)$ can be read from row totals of the above table. That is, $f_1(0)=f(0,0)+f(0,1)+f(0,2)=\frac{1}{36}+\frac{6}{36}+\frac{3}{36}=\frac{10}{36}$. Similarly, $f_1(1)=\frac{20}{36}$ $f_1(2)=\frac{6}{36}$ and $f_1(x)=0$ for any other values of x. The marginal p.f. of X gives the probabilities of the number of white balls in the sample. Similarly, the column totals represent the marginal p.f. of Y.

The marginal distributions derived above may also be displayed in tabular form as below.

Marginal distribution of X :

x	0	1	2
$f_1(x)$	$\frac{10}{36}$	$\frac{20}{36}$	$\frac{6}{36}$

Marginal distribution of Y :

y	0	1	2
$f_1(y)$	$\frac{10}{36}$	$\frac{20}{36}$	$\frac{6}{36}$

Note that total probability in each of the above marginal distributions is 1. This shows that marginal distributions are necessarily probability distributions.

The marginal p.f. can also be derived from the joint p.d.f.

$$f(x,y)=P(X=x,Y=y)=\begin{cases}\dfrac{\binom{4}{x}\binom{3}{y}\binom{2}{2-x-y}}{\binom{9}{2}} & \text{for } x=0,1,2;\, y=0,1,2;\, 0\le x+y\le 2,\\[2ex] 0 & \text{elsewhere.}\end{cases}$$

For $x=0,1,2$, marginal p.f. of X is

$$\sum_{\text{All } y} f(x,y)=\sum_{y=0}^{2}\frac{\binom{4}{x}\binom{3}{y}\binom{2}{2-x-y}}{\binom{9}{2}}$$

$$=\frac{\binom{4}{x}}{\binom{9}{2}}\sum_{y=0}^{2}\binom{3}{y}\binom{2}{2-x-y}$$

$$=\frac{\binom{4}{x}\binom{5}{2-x}}{\binom{9}{2}}$$

Thus, the marginal p.f. of X is

$$f_1(x)=\begin{cases}\dfrac{\binom{4}{x}\binom{5}{2-x}}{\binom{9}{2}} & \text{for } x=0,1,2,\\ 0 & \text{elsewhere.}\end{cases}$$

For $y=0,1,2$, the marginal p.f. of Y is

$$\sum_{\text{All } x} f(x,y)=\sum_{x=0}^{2}\frac{\binom{4}{x}\binom{3}{y}\binom{2}{2-x-y}}{\binom{9}{2}}$$

$$=\frac{\binom{3}{y}}{\binom{9}{2}}\sum_{x=0}^{2}\binom{4}{x}\binom{2}{2-x-y}$$

$$=\frac{\binom{3}{y}\binom{6}{2-y}}{\binom{9}{2}}$$

Thus, the marginal p.f. of Y is

$$f_2(y)=\begin{cases}\dfrac{\binom{3}{y}\binom{6}{2-y}}{\binom{9}{2}} & \text{for } y=0,1,2,\\ 0 & \text{elsewhere.}\end{cases}$$

Example 11.28 The joint probability function of a discrete bivariate random variable

$$f(x,y)=\begin{cases}\dfrac{1}{36} & \text{for } 1\le x=y\le 6,\\ \dfrac{2}{36} & \text{for } 1\le y<x\le 6,\\ 0 & \text{elsewhere.}\end{cases}$$

Find the marginal p.f.s of X and Y.

Solution The marginal p.f. $f_1(x)$ of X can be obtained by summing the joint p.f. $f(x,y)$ for all y values of the random variable Y. That is,

for $x=1,2,\cdots,6$, the marginal p.f. of X is

$$f_1(x)=\sum_{\text{All } y} f(x,y)$$

$$= \sum_{y=1}^{6} f(x,y)$$

$$= f(x,x) + \sum_{y<x}^{6} f(x,y) + \sum_{y>x}^{6} f(x,y)$$

$$= \frac{1}{36} + (x-1)\frac{2}{36} + 0$$

$$= \frac{2x-1}{36}.$$

Thus,

$$f_1(x) = \begin{cases} \dfrac{2x-1}{36} & \text{for } x = 1, 2, \cdots, 6, \\ 0 & \text{elsewhere.} \end{cases}$$

For $y = 1, 2, \cdots, 6$, the marginal p.f. of Y is

$$f_2(y) = \sum_{\text{All } x} f(x,y)$$

$$= \sum_{x=1}^{6} f(x,y)$$

$$= f(y,y) + \sum_{x<y}^{6} f(x,y) + \sum_{x>y}^{6} f(x,y)$$

$$= \frac{1}{36} + 0 + (6-y)\frac{2}{36}$$

$$= \frac{13-2y}{36}.$$

Thus,

$$f_2(y) = \begin{cases} \dfrac{13-2y}{36} & \text{for } y = 1, 2, \cdots, 6, \\ 0 & \text{elsewhere.} \end{cases}$$

Note that if $f(x,y)$ is a joint probability density function of a continuous bivariate random variable (X,Y), then the marginal p.d.f. $f_1(x)$ of X is obtained (replacing the sum over all possible values of Y) by the integral over all possible values of Y.

Definition 11.10.3 Let $f(x,y)$ be a joint probability density function of a continuous bivariate random variable (X,Y). Then the **marginal probability density function** (marginal p.d.f.) of X is

$$f_1(x) \text{ or } g(x) = \int_{-\infty}^{\infty} f(x,y)dy \quad \text{for } -\infty < y < \infty.$$

Similarly, the marginal p.d.f. of Y is

$$f_2(y) \text{ or } h(y) = \int_{-\infty}^{\infty} f(x,y)dx \quad \text{for } -\infty < x < \infty.$$

Example 11.29 Determining a Marginal p.d.f. With reference to Example 11.23, find the marginal p.d.f.s of X and Y. The joint p.d.f. of X and Y is

$$f(x,y)=\begin{cases}\frac{6}{5}\left(x^2+2xy\right) & \text{if } 0\le x\le 1,\ 0\le y\le 1,\\ 0 & \text{otherwise.}\end{cases}$$

Solution It is seen that X cannot take any value outside the interval $[0,1]$. Therefore, for $0\le x\le 1$, the marginal p.d.f. of X is

$$f_1(x)=\int_{-\infty}^{\infty} f(x,y)\,dy$$

$$=\frac{6}{5}\int_0^1 \left(x^2+2xy\right)dy$$

$$=\frac{6}{5}\left[x^2y+xy^2\right]_0^1=\frac{6}{5}x(x+1).$$

Thus, the marginal p.d.f. of X is

$$f_1(x)=\begin{cases}\frac{6}{5}x(x+1) & \text{for } 0\le x\le 1,\\ 0 & \text{elsewhere.}\end{cases}$$

For $0\le y\le 1$, the marginal p.d.f. of Y is

$$f_2(y)=\int_{-\infty}^{\infty} f(x,y)\,dx$$

$$=\frac{6}{5}\int_0^1 \left(x^2+2xy\right)dx$$

$$=\frac{6}{5}\left[\frac{x^3}{3}+x^2y\right]_0^1=\frac{6}{5}\left(\frac{1}{3}+y\right)=\frac{2}{5}(1+3y).$$

Thus, the marginal p.d.f. of Y is

$$f_2(y)=\begin{cases}\frac{2}{5}(1+3y) & \text{for } 0\le y\le 1,\\ 0 & \text{elsewhere.}\end{cases}$$

Example 11.30 Determining a Marginal p.d.f. Suppose that the joint p.d.f. a bivariate random variable (X,Y) is given by

$$f(x,y)=\begin{cases}\frac{21}{4}xy^2 & \text{for } y^2\le x\le 1,\\ 0 & \text{elsewhere.}\end{cases}$$

Find marginal densities of X and Y.

Solution The domain of the function consists of the region bounded by the curve $x=y^2$ and the horizontal line $x=1$ (See the Figure 11.10.2).

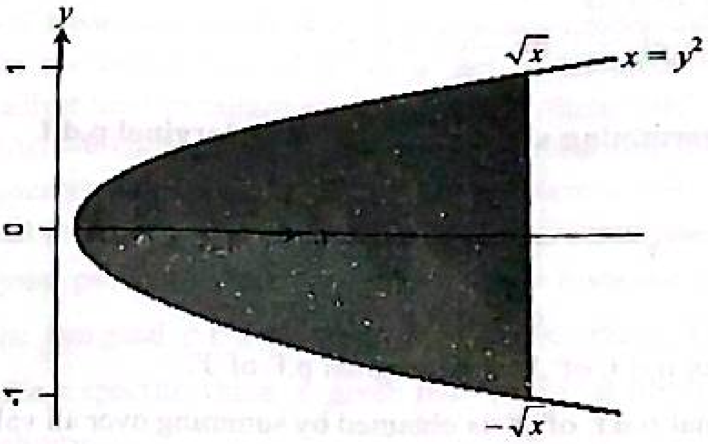


Figure 11.10.2 $f(x, y) > 0$ **in the shaded region**

X cannot take any value outside the interval $[0,1]$. Therefore, $f(x,y)=0$ outside the interval. Also, $f(x,y)=0$ unless $-\sqrt{x} \le y \le \sqrt{x}$. Therefore, for $0 \le x \le 1$,

$$f_1(x)=\int_{-\infty}^{\infty} f(x,y)dy=\int_{-\sqrt{x}}^{\sqrt{x}} \frac{21}{4}xy^2\,dy=\frac{21}{4}x\left[\frac{y^3}{3}\right]_{-\sqrt{x}}^{\sqrt{x}}=\frac{7}{4}x\left[\left(\sqrt{x}\right)^3-\left(-\sqrt{x}\right)^3\right]=\frac{7}{2}x^{5/2}.$$

Thus, the marginal p.d.f. of X is

$$f_1(x)=\begin{cases}\dfrac{7}{2}x^{5/2} & \text{for } 0 \le x \le 1,\\ 0 & \text{elsewhere.}\end{cases}$$

Again, Y cannot take any value outside the interval $[-1,1]$. Therefore, $f(x,y)=0$ outside the interval. Also, $f(x,y)=0$ unless $y^2 \le x \le 1$. Thus, for $-1<y<1$,

$$f_2(y)=\int_{-\infty}^{\infty} f(x,y)dx=\int_{y^2}^{1} \frac{21}{4}xy^2\,dx=\frac{21}{4}y^2\left[\frac{x^2}{2}\right]_{y^2}^{1}=\frac{21}{4}y^2\,\frac{1-y^4}{2}$$

Thus, the marginal p.d.f. of Y is

$$f_2(y)=\begin{cases}\dfrac{21}{8}y^2\left(1-y^4\right) & \text{for} -1<y<1,\\ 0 & \text{elsewhere.}\end{cases}$$

If $f(x,y)$ is a joint probability distribution of a bivariate random variable (X,Y), where X is discrete and Y is continuous, then we can derive the marginal p.f. of X and the marginal p.d.f. of Y in the same way that we derived a marginal p.f. or a marginal p.d.f. from a joint p.f. or a joint p.d.f. Let $f(x,y)$ be the joint p.f./p.d.f. of discrete random variable X and continuous random variable Y. Then the marginal p.f. of X is

$$f_1(x)=\int_{-\infty}^{\infty} f(x,y)dy, \qquad \text{for all } x,$$

and the marginal p.d.f. of Y is

$$f_2(y)=\sum_{\text{All } x} f(x,y), \qquad \text{for } -\infty \le x \le \infty.$$

Example 11.31 Determining a Marginal p.f. and Marginal p.d.f. Suppose the joint p.f./p.d.f. of X and Y is

$$f(x,y)=\begin{cases} \frac{1}{4}x^{y-1}y & \text{for } 0<x<1 \text{ and } y=1,2,3,4, \\ 0 & \text{elsewhere.} \end{cases}$$

Determine the marginal p.d.f. of X and marginal p.f. of Y.

Solution The marginal p.d.f. of X is obtained by summing over all values of Y as

$$f_1(x)=\sum_{\text{All } y} f(x,y)=\sum_{y=1}^{4}\frac{1}{4}x^{y-1}y=\frac{1}{4}\left(1+2x+3x^2+4x^3\right)$$

Thus,

$$f_1(x)=\begin{cases} \frac{1}{4}\left(1+2x+3x^2+4x^3\right) & \text{for } 0<x<1, \\ 0 & \text{elsewhere.} \end{cases}$$

The marginal p.f. of Y is obtained by integrating over the range of X as

$$f_2(y)=\int_{-\infty}^{\infty} f(x,y)dx=\int_0^1 \frac{1}{4}x^{y-1}y\,dx=\frac{1}{4}\left[x^y\right]_0^1=\frac{1}{4}.$$

Thus,

$$f_2(y)=\begin{cases} \frac{1}{4} & \text{for } y=1,2,3,4, \\ 0 & \text{elsewhere.} \end{cases}$$

11.11 Conditional Distributions

We extend the concept of conditional probability to conditional distributions. The conditional distribution of one random variable X given another random variable Y will be the distribution that we would use for X after knowing the value of Y.

11.11.1 Discrete Conditional Distributions

Example 11.32 Auto Insurance. A life insurance company is interested to keep track of how likely smoking contributes lung cancer. A joint probability distribution of smoking and lung cancer is given below:

Table 11.6 Joint probability of Smoking and Lung caner

	Smoking Y		
Cancer X	0	1	Total
0	0.66	0.17	0.83
1	0.12	0.05	0.17
Total	0.78	0.22	1.00

We let $X=1$ if a person has cancer and 0 if not. We let $Y=1$ if the person smokes and 0 if not. The probability that a person will have lung cancer is 0.17. If a policy holder applies for insurance, the company needs to compute the distribution of X as a part of its premium determination. The insurance company might adjust their premium according to risk factor such as likelihood of being lung cancer. If we assume that we know the smoking habit, the probability might change quite a bit. We introduce the formal concepts for addressing this type of problem in this section.

The notion of conditional probability can easily be extended to conditional probability function. Suppose that $f(x,y)$ is a joint probability function of a discrete bivariate random variable (X,Y). Let $f_1(x)$ and $f_2(y)$ be the marginal p.f.'s of X and Y, respectively. The probability that the random variable X will take a specific value x given that $Y=y$ is observed is specified by the following conditional probability:

$$P(X=x\mid Y=y)=\frac{P(X=x,Y=y)}{P(Y=y)}=\frac{f(x,y)}{f_2(y)}. \qquad \text{(i)}$$

If we know that $Y=y$ is observed, then we can update the probability of $X=x$ by the above equation (i). Finally, we consider the entire distribution of X after learning that $Y=y$.

Definition 11.11.1 Let $f(x,y)$ be a discrete joint probability function of a discrete bivariate random variable (X,Y). Let $f_2(y)$ denote the marginal probability function of Y. For any y of Y such that $f_2(y)=P(Y=y)>0$, the **conditional probability function** of X, given $Y=y$ is defined as

$$g(x\mid y)=\frac{f(x,y)}{f_2(y)} \quad \text{for all } x.$$

Similarly, if $f_1(x)$ denotes the marginal probability function of X, then for any x of X such that $f_1(x)=P(X=x)>0$, the conditional probability function of Y, given $X=x$ is defined as

$$h(y\mid x)=\frac{f(x,y)}{f_1(x)} \quad \text{for all } y.$$

We should verify that $g(x\mid y)$ is a probability function for each y such that $f_2(y)>0$. It is clear that $g(x\mid y)\geq 0$ for all x and

$$\sum_x g(x\mid y)=\frac{1}{f_2(y)}\sum_x f(x,y)=\frac{1}{f_2(y)}.f_2(y)=1.$$

Example 11.33 Determining conditional p.f. from joint p.f. Suppose that the joint p.f. of X and Y is as specified in Table 11.6 in Example 11.32. We shall determine the conditional p.f. of X given $Y=1$.

The marginal probability function of Y appears in the row total of Table 11.6, so that $f_2(1)=P(Y=1)=0.22$. Therefore, the conditional probability $g(x\mid 1)$ that X will take a particular value x is

$$g(x\mid 1)=\frac{f(x,1)}{f_2(1)}=\frac{f(x,1)}{0.22}.$$

Therefore, the conditional probability distribution of X given $Y = 1$ is

$$g(x \mid 1) = \frac{f(x,1)}{0.22}, \text{ for } x = 0, 1.$$

Therefore,

$$g(0 \mid 1) = \frac{f(0,1)}{0.22} = \frac{0.17}{0.22} = 0.773.$$

$$g(1 \mid 1) = \frac{f(1,1)}{0.22} = \frac{0.05}{0.22} = 0.227.$$

We shall also determine the conditional p.f. of X given Y.

The conditional distribution of X (being cancer) given Y (Smoking habit) is given in Table 11.7.

Table 11.7 Conditional p.f. of X given Y

	Smoking Y	
Cancer X	0	1
0	0.846	0.773
1	0.154	0.227

It appears that a smoker is much more likely to be a lung cancer patient. That is, a smoker has a significant chance of being cancer.

Example 11.34 Determining conditional p.f. from joint p.f. With reference to the joint probability function of Example 11.27, determine the conditional probability function of Y given $X = 1$.

Solution The marginal probability function of X appears in the column total of Table 11.4 of Example 11.22 on page 502. So, $f_1(1) = P(X = 1) = \frac{20}{36}$. Therefore, the conditional probability $h(y \mid x)$ that Y will take a specific value y is

$$h(y \mid 1) = \frac{f(1,y)}{f_1(1)} = \frac{f(1,y)}{20/36}.$$

Thus, the conditional distribution of Y given $X = 1$ is

$$h(y \mid 1) = \frac{f(1,y)}{20/36} \quad \text{for } y = 0, 1, 2.$$

Therefore,

$$h(0 \mid 1) = \frac{f(1,0)}{20/36} = \frac{8/36}{20/36} = \frac{2}{5}$$

$$h(1 \mid 1) = \frac{f(1,1)}{20/36} = \frac{12/36}{20/36} = \frac{3}{5}$$

$$h(2 \mid 1) = \frac{f(1,2)}{20/36} = \frac{0}{20/36} = 0$$

Hence the conditional probability distribution of Y given $X = 1$ in tabular form is

y	0	1	2
$h(y \mid 1)$	$\frac{2}{5}$	$\frac{3}{5}$	0

Example 11.35 Determining conditional p.f. from joint p.f. With reference to the joint probability function of Example 11.28, determine the conditional probability function of X given $Y = 5$.

Solution The joint p.f. of (X, Y) is

$$f(x,y)=\begin{cases}\frac{1}{36} & \text{for } 1 \le x = y \le 6,\\ \frac{2}{36} & \text{for } 1 \le y < x \le 6,\\ 0 & \text{elsewhere.}\end{cases}$$

and

$$f_2(y)=\begin{cases}\frac{13-2y}{36} & \text{for } y = 1, 2, \cdots, 6,\\ 0 & \text{elsewhere.}\end{cases}$$

We want to find

$$g(x \mid 5)=\frac{f(x,5)}{f_2(5)}, \text{ for } x = 5, 6.$$

Now, $f(x,5)=\begin{cases}\frac{1}{36}, & \text{for } x = 5,\\ \frac{2}{36}, & \text{for } x = 6.\end{cases}$

Now, $f_2(5)=\dfrac{13-(2)(5)}{36}=\dfrac{3}{36}.$

Hence,

$$g(x \mid 5)=\begin{cases}\frac{1}{3} & \text{for } x = 5,\\ \frac{2}{3} & \text{for } x = 6,\\ 0 & \text{elsewhere.}\end{cases}$$

11.11.2 Continuous Conditional Distributions

If a continuous distribution is calculated conditionally on some information, then the density is called a conditional density. When the conditioning information involves another random variable with a continuous distribution, the conditional density can be calculated from the joint density for the two random variables. In other words, suppose that X and Y has a joint continuous distribution. We would like to be able to compute a conditional distribution for X given $Y = y$. Since $\{Y = y\}$ is an event with probability 0 for all y, conditional p.d.f. cannot be determined in the same way as condtional discrete p.f.

Definition 11.11.2 Let $f(x,y)$ be a joint probability density function of a continuous bivariate random variable (X,Y). Let $f_2(y)$ denote the marginal probability function of Y. Let y be a value such that $f_2(y)>0$. Then the conditional probability function of X, given $Y=y$, is defined as

$$g(x|y)=\frac{f(x,y)}{f_2(y)}, \qquad \text{for } -\infty<x<\infty.$$

For values of y such that $f_2(y)=0$, then $g(x|y)$ is defined to be any p.d.f. we wish.

Note that formulae of conditional p.f. for discrete random variable and conditional p.d.f. for continuous random variable are identical. However, conditional p.f. for discrete random variable is derived as the conditional probability that $X=x$ given that $Y=y$, whereas, conditional p.d.f. for continuous random variable is defined to be the value of the conditional p.d.f. of X given that $Y=y$.

For each y, $g(x|y)\geq 0$. Also, if $f_2(y)>0$,

$$\int_{-\infty}^{\infty} g(x|y)dx=\frac{\int_{-\infty}^{\infty} f(x,y)dx}{f_2(y)}=\frac{f_2(y)}{f_2(y)}=1.$$

Similarly, for each value of x such that $f_1(x)>0$, the conditional p.d.f. of Y, given $X=x$, is defined as

$$h(y|x)=\frac{f(x,y)}{f_1(x)}, \qquad \text{for } -\infty<y<\infty.$$

For values of x such that $f_1(x)=0$, then $h(y|x)$ is defined to be any p.d.f. we wish.

Example 11.36 Calculating Conditional p.d.f. form joint p.d.f. With reference to Example 11.29, determine the conditional distribution of Y given $X=x$, the conditional distribution of Y given $X=\frac{1}{2}$ and then determine $P\left(Y>\frac{3}{4}\middle|X=\frac{1}{2}\right)$.

Solution The joint p.d.f. of (X,Y) is

$$f(x,y)=\begin{cases}\frac{6}{5}\left(x^2+2xy\right) & \text{for } 0\leq x\leq 1,\ 0\leq y\leq 1,\\ 0 & \text{elsewhere.}\end{cases}$$

The marginal p.d.f. of X is

$$f_1(x)=\begin{cases}\frac{6}{5}x(x+1) & \text{if } 0\leq x\leq 1,\\ 0 & \text{otherwise.}\end{cases}$$

For each $0\leq x\leq 1$, the conditional p.d.f. of Y given $X=x$ is

$$h(y|x)=\frac{f(x,y)}{f_1(x)}=\frac{\frac{6}{5}\left(x^2+2xy\right)}{\frac{6}{5}x(x+1)}=\frac{x+2y}{x+1}, \text{ for } 0\leq y\leq 1.$$

Thus,

$$h(y|x)=\begin{cases}\dfrac{x+2y}{x+1} & \text{for } 0\le y\le 1,\\ 0 & \text{elsewhere.}\end{cases}$$

The conditional p.d.f. of Y given $X=\frac{1}{2}$ (which lies between 0 and 1) is

$$h\left(y|X=\frac{1}{2}\right)=\begin{cases}\dfrac{\frac{1}{2}+2y}{\frac{1}{2}+1} & \text{for } 0\le y\le 1,\\ 0 & \text{elsewhere.}\end{cases}$$

$$=\begin{cases}\dfrac{1+4y}{3} & \text{for } 0\le y\le 1,\\ 0 & \text{elsewhere.}\end{cases}$$

$$P\left(Y>\frac{3}{4}\middle|X=\frac{1}{2}\right)=\int_{3/4}^{1} h\left(y\middle|\frac{1}{2}\right)dy$$

$$=\int_{3/4}^{1}\frac{1+4y}{3}dy=\frac{1}{3}\left[y+2y^2\right]_{3/4}^{1}=\frac{1}{3}\left[3-\left(\frac{3}{4}+2\frac{9}{16}\right)\right]=\frac{1}{3}\left[3-\frac{15}{8}\right]=\frac{3}{8}.$$

Example 11.37 Calculating Conditional p.d.f. from joint p.d.f. With reference to Example 11.30, determine the conditional distribution of X given $Y=y$, and then determine $P\left(X<\frac{3}{4}\middle|Y=\frac{1}{2}\right)$.

Solution The joint p.d.f. of (X,Y) is

$$f(x,y)=\begin{cases}\dfrac{21}{4}xy^2 & \text{for } y^2\le x\le 1,\\ 0 & \text{elsewhere.}\end{cases}$$

The marginal p.d.f. of Y is

$$f_2(y)=\begin{cases}\dfrac{21}{8}y^2\left(1-y^4\right) & \text{for } -1<y<1,\\ 0 & \text{otherwise.}\end{cases}$$

It is clear that $f_2(y)>0$ for $-1<y<1$, and $f_2(y)=0$, for $y=0$. Therefore, for each given value of y such that $-1<y<0$ or $0<y<1$, the conditional p.d.f. $g(x\mid y)$ of X is

$$g(x\mid y)=\begin{cases}\dfrac{f(x,y)}{f_2(y)}, & \text{for } y^2\le x\le 1,\\ 0, & \text{otherwise,}\end{cases}$$

$$= \begin{cases} \dfrac{\frac{21}{4}xy^2}{\frac{21}{8}y^2\left(1-y^4\right)} & \text{for } y^2 \le x \le 1, \\ 0 & \text{elsewhere,} \end{cases}$$

$$= \begin{cases} \dfrac{2x}{1-y^4} & \text{for } y^2 \le x \le 1, \\ 0 & \text{elsewhere.} \end{cases}$$

If it is known that $Y = \frac{1}{2}$, then $P\left(\frac{1}{4} \le X \le 1 \middle| Y = \frac{1}{2}\right) = 1$, and

$$P\left(X < \frac{3}{4} \middle| Y = \frac{1}{2}\right) = \int_{1/4}^{3/4} g\left(x \middle| \frac{1}{2}\right) dx$$

$$= \int_{1/4}^{3/4} \frac{2x}{1-(1/2)^2} dx = \frac{4}{3}\left[x^2\right]_{1/4}^{3/4} = \frac{4}{3}\left[\left(\frac{3}{4}\right)^2 - \left(\frac{1}{4}\right)^2\right] = \left(\frac{4}{3}\right)\left(\frac{8}{16}\right) = \frac{2}{3}.$$

Example 11.38 Calculating Conditional p.d.f. from joint p.d.f. Let X and Y be continuous random variables with joint p.d.f.

$$f(x, y) = \begin{cases} 12x & \text{for } 0 < y < 2x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Determine the conditional distribution of Y given $X = x$. Also determine $P\left(Y > \frac{1}{8} \middle| X = \frac{1}{2}\right)$.

Solution The marginal distribution of X is

$$f_1(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^{2x} 12x dy = 12x[y]_0^{2x} = 24x^2.$$

Thus, $f_1(x) = \begin{cases} 24x^2 & \text{for } 0 \le x \le \frac{1}{2}, \\ 0 & \text{elsewhere.} \end{cases}$

It is clear $f_1(x) > 0$ for each value of $0 < x \le \frac{1}{2}$, and 0 otherwise. Thus for each given value of x such that $0 < x \le \frac{1}{2}$, the conditional p.d.f. $h(y \mid x)$ of Y is

$$h(y \mid x) = \begin{cases} \dfrac{f(x, y)}{f_1(x)} & \text{for } 0 < y \le 2x, \\ 0 & \text{elsewhere,} \end{cases}$$

$$= \begin{cases} \dfrac{12x}{24x^2} & \text{for } 0 < y \le 2x, \\ 0 & \text{otherwise,} \end{cases}$$

$$= \begin{cases} \frac{1}{2x} & \text{for } 0 < y \le 2x, \\ 0 & \text{elsewhere.} \end{cases}$$

$$p\left(Y > \frac{1}{8} \middle| X = \frac{1}{2}\right) = \int_{1/8}^{1} 1dy = [y]_{1/8}^{1} = 1 - \frac{1}{8} = \frac{7}{8}.$$

11.12 Independent Random Variables

We know that two events A and B are said to be independent if $P(A \cap B) = P(A)P(B)$. Let (X,Y) denote the joint random variable. Let us define two events as $A = \{X \le x\}$ and $B = \{Y \le y\}$. Thus, $P(A \cap B) = P(X \le x, Y \le y)$. Therefore, the random variables X and Y are said to be independent if

$$P(X \le x, Y \le y) = P(X \le x)P(Y \le y)$$

which implies

$$F(x,y) = F_1(x)F_2(y), \; -\infty < x < \infty \text{ and } -\infty < y < \infty.$$

From relationship of c.f. and p.f. (c.d.f. and p.d.f.), we have

$$f(x,y) = f_1(x)f_2(y), \; -\infty < x < \infty \text{ and } -\infty < y < \infty.$$

Definition 11.12.1 Let the joint c.d.f. of a bivariate random variable (X,Y) be $F(x,y)$ and the marginal c.d.f. of X be $F_1(x)$ and the marginal c.d.f. be $F_2(y)$. Then X and Y are independent if and only if, for all real numbers x and y,

$$F(x,y) = F_1(x)F_2(y).$$

In other words, let the joint p.f./p.d.f. of a bivariate random variable (X,Y) be $f(x,y)$ and the marginal p.f./p.d.f. of X be $f_1(x)$ and the marginal p.f/p.d.f. of Y be $f_2(y)$. Then X and Y are independent if and only if, for all real numbers x and y,

$$f(x,y) = f_1(x)f_2(y).$$

Example 11.39 Demand for Utilities. With reference to Example 11.26, the joint c.d.f for water and electricity demand (X,Y) is

$$F(x,y) = \begin{cases} 0 & \text{for } x < 4 \text{ or } y < 2, \\ \frac{(x-4)(y-2)}{19208} & \text{for } 4 \le x < 200 \text{ and } 2 \le y < 100, \\ \frac{x-4}{196} & \text{for } 4 \le x < 200 \text{ and } y \ge 100, \\ \frac{y-2}{98} & \text{for } x \ge 200 \text{ and } 2 \le y < 100, \\ 1 & \text{for } x \ge 200 \text{ and } y \ge 100. \end{cases}$$

The marginal c.d.f. for water demand is

$$F_1(x)=\begin{cases}0 & \text{for } x<4,\\ \frac{x-4}{196} & \text{for } 4\le x<200,\\ 1 & \text{for } x>200,\end{cases}$$

and the marginal c.d.f. for electricity demand is

$$F_2(y)=\begin{cases}0 & \text{for } y<2,\\ \frac{y-2}{98} & \text{for } 2\le y<100,\\ 1 & \text{for } y\ge 100.\end{cases}$$

Clearly, the product of the marginal p.d.f.'s of water and electric demand equals their joint p.d.f. Thus, the two random variables 'water demand and electricity demand' are independent. The independence of X and Y can be verified using density function as follows.

The joint p.d.f for water and electricity demand (X,Y) is (Ref. to Example 11.39)

$$f(x,y)=\begin{cases}\frac{1}{19208} & \text{for } 4\le x\le 200 \text{ and } 2\le y\le 100,\\ 0 & \text{elsewhere.}\end{cases}$$

The marginal p.d.f.s of X and Y are (Ref. to section 11.39)

$$f_1(x)=\begin{cases}\frac{1}{196} & \text{if } 4\le x\le 200,\\ 0 & \text{elsewhere,}\end{cases}$$

and

$$f_2(y)=\begin{cases}\frac{1}{98} & \text{for } 2\le y\le 100,\\ 0 & \text{elsewhere.}\end{cases}$$

It is clear that $f_1(x)f_2(y)=\frac{1}{196}\cdot\frac{1}{98}=\frac{1}{19208}=f(x,y)$ for $4\le x\le 200$ and $2\le y\le 100$,

$=0$, otherwise.

Hence, $f(x,y)=f_1(x)f_2(y)$, for $-\infty<x<\infty$ and $-\infty<y<\infty$.

Thus, the two random variables 'water demand and electricity demand' are independent.

Example 11.40 Consider the experiment of tossing a fair coin and a fair die simultaneously. Let X stand for the number of heads on the coin toss, and let Y stand for the number of spots on the die. Are X and Y independent?

Solution The joint probability function of X and Y is

$$f(x,y)=\begin{cases}\frac{1}{12} & \text{for } x=0,1 \text{ and } y=1,2,3,4,5,6,\\ 0 & \text{elsewhere.}\end{cases}$$

The marginal probability function of X is

$$f_1(x) = \sum_{\text{all } y} f(x, y) = \begin{cases} \sum_{y=1}^{6} \frac{1}{12} & \text{for } x = 1, 2, \\ 0 & \text{elsewhere}, \end{cases}$$

$$= \begin{cases} \frac{1}{2} & \text{for } x = 0, 1, \\ 0 & \text{elsewhere}. \end{cases}$$

The marginal probability function of Y is

$$f_2(y) = \sum_{\text{all } x} f(x, y) = \begin{cases} \sum_{x=1}^{2} \frac{1}{12} & \text{for } y = 1, 2, 3, 4, 5, 6, \\ 0 & \text{elsewhere}, \end{cases}$$

$$= \begin{cases} \frac{1}{6} & \text{for } y = 1, 2, 3, 4, 5, 6, \\ 0 & \text{elsewhere}. \end{cases}$$

Now, $f_1(x)f_2(y) = \frac{1}{2}.\frac{1}{6} = \frac{1}{12} = f(x, y)$, for $x = 1, 2$ and $y = 1, 2, 3, 4, 5, 6$,

$= 0$ elsewhere.

Hence, $f(x, y) = f_1(x)f_2(y)$, for all real x and y.

Thus, the two random variables X and Y are independent.

Displaying the joint p.f. $f(x, y)$ in tabular form, the independence of X and Y can be checked in the following way.

x	y = 1	2	3	4	5	6	Row total
0	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{2}$
1	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{2}$
Column total	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	1

The marginal p.f. $f_1(x)$ is shown in the row total and the marginal p.f. $f_2(y)$ is shown in the column total.

Now, $f_1(0)f_2(1) = P(X = 0)P(Y = 1) = \frac{1}{2}.\frac{1}{6} = \frac{1}{12} = f(0, 1)$

$f_1(0)f_2(2) = P(X = 0)P(Y = 2) = \frac{1}{2}.\frac{1}{6} = \frac{1}{12} = f(0, 2)$

$\vdots$

$f_1(1)f_2(6) = P(X = 1)P(Y = 6) = \frac{1}{2}.\frac{1}{6} = \frac{1}{12} = f(1, 6)$

That is, the above computations show that for all $x = 0, 1$ and $y = 0, 1, 2, 3, 4, 5, 6$, we have

$$f(x, y) = f_1(x) f_2(y).$$

Elsewhere, $f(x, y) = 0 = f_1(x) f_2(y)$.

Hence, the two random variables X and Y are independent.

The meaning of independence Two events are said to be independent if learning that one of them occurs does not change the probability that the other one occurs. It is easiest to extend this idea to discrete random variables. Suppose that X and Y have a discrete probability function. If for each y, learning that $Y = y$ does not change any of the probabilities of the events $\{X = x\}$, we would like to say that X and Y are independent. Thus, X and Y are said to be independent if and only if for every value of y such that $f_2(y) > 0$, and for every value of x

$$g(x \mid y) = f_1(x).$$

Similarly, X and Y are said to be independent if and only if for every value of x such that $f_1(x) > 0$, and for every value of y

$$h(y \mid x) = f_2(y).$$

For Example 11.40, for each $y = 1, 2, 3, 4, 5, 6$, the conditional distribution of X given $Y = y$ is

$$g(x \mid y) = \frac{f(x, y)}{f_2(y)} = \frac{1/12}{1/6} = \frac{1}{2} = f_1(x) \text{ for } x = 0, 1.$$

This establishes the independence of X and Y.

Example 11.41 Suppose that the joint p.d.f. of X and Y has the following form:

$$f(x, y) = \begin{cases} 24xy, & \text{for } x \geq 0, y \geq 0 \text{ and } x + y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Are X and Y independent?

Solution

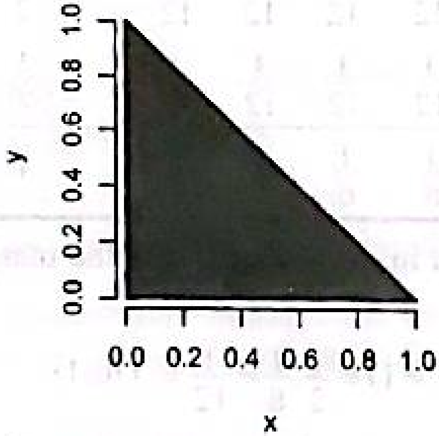


Figure 11.12.1. $f(x,y) > 0$ **in the shaded region**

The marginal distribution of X for $0 \leq x \leq 1$ is

$$f_1(x) = \int_{-\infty}^{\infty} f(x, y) dy = 24x \int_0^{1-x} y dy = 12x\left[y^2\right]_0^{1-x} = 12x(1-x)^2.$$

Thus, $f_1(x)=\begin{cases}12x(1-x)^2 & \text{for } 0\le x\le 1,\\ 0 & \text{otherwise.}\end{cases}$

Similarly, the marginal distribution of Y is

$$f_2(y)=\begin{cases}12y(1-y)^2 & \text{for } 0\le y\le 1,\\ 0 & \text{otherwise.}\end{cases}$$

Thus, $f(x,y)\neq f_1(x)f_2(y)$ for $x\ge 0, y\ge 0$, and $x+y\le 1$.

Hence, X and Y are not independent.

For example, $f(0.9,0.9)=0$, but neither $f_1(0.9)=0$ nor $f_2(0.9)=0$.

The important feature of this example is that the values of X and Y are constrained to lie inside a line bounded by axes. In this region, the joint p.d.f. of X and Y is positive and zero outside the region. Under this conditions X and Y cannot be independent, because for every given value x of X, the possible values of Y depend on x. For example, if $X=0$, then Y can have any value such that $Y\le 1$; if $X=1/4$, then Y must have a value such that $Y\le 3/4$.

Example 11.42 With reference to the joint probability function of Example 11.27, check whether the random variable X and Y are independent.

Solution

Joint and marginal probability functions of (X,Y)

x	y = 0	1	2	Row total
0	$\frac{1}{36}$	$\frac{6}{36}$	$\frac{3}{36}$	$\frac{10}{36}$
1	$\frac{8}{36}$	$\frac{12}{36}$	0	$\frac{20}{36}$
2	$\frac{6}{36}$	0	0	$\frac{6}{36}$
Column total	$\frac{15}{36}$	$\frac{18}{36}$	$\frac{3}{36}$	1

The row total and column total show the marginal distributions of X and Y respectively.

Now, $f(0,0)=\frac{1}{36}$, $f_1(0)=P(X=0)=\frac{10}{36}$, and $f_2(0)=P(Y=0)=\frac{15}{36}$.

$f_1(0)f_2(0)=\frac{10}{36}\cdot\frac{15}{36}=\frac{25}{216}\neq\frac{1}{36}=f(0,0)$.

Hence, X and Y are not independent.

11.13 Multiplication Rule for Conditional Distributions

If A and B are two events, then we have $P(A\cap B)=P(A)P(B\mid A)$. This multiplication rule for conditional probabilities can be generalized to the case of two random variables.

Let X and Y be random variables such that X has p.f. (or p.d.f.) $f_1(x)$ and Y has p.f. (or p.d.f.) $f_2(y)$. Assume that the conditional p.f. (or p.d.f.) of X given $Y = y$ is $g(x \mid y)$. The conditional p.f. (or p.d.f.) of Y given $X = x$ is $h(y \mid x)$. Then for each y such that $f_2(y) > 0$ and for each x,

$$f(x, y) = g(x \mid y) f_2(y).$$

Similarly for each x such that $f_1(x) > 0$, and for each y,

$$f(x, y) = h(y \mid x) f_1(x).$$

If $f_2(y_0) = 0$ for some value y_0 of Y. Then, without loss of generality, it can be assumed that $f(x, y_0) = 0$ for all values x of X. In this case, both sides will be zero. Hence, for all real values of x and y,

$$f(x, y) = g(x \mid y) f_2(y) = h(y \mid x) f_1(x).$$

Example 11.43 Let X be the rate (calls per hour) at which calls arrive at a switch board. Let Y be the number of calls during a three-hour period. Suppose that the marginal p.d.f. of X is

$$f_1(x) = \begin{cases} e^{-x} & \text{for } x > 0, \\ 0 & \text{elsewhere,} \end{cases}$$

and that the conditional p.f. of Y given $X = x$ is

$$h(y \mid x) = \begin{cases} \dfrac{e^{-3x}(3x)^y}{y!} & \text{for } y = 0, 1, 2, \cdots, \\ 0 & \text{elsewhere.} \end{cases}$$

a. Find the joint p.f. of (X, Y) and marginal p.f. of Y.

b. Find the conditional p.d.f. $g(x \mid 0)$ of X given $Y = 0$.

Solution The joint distribution of X and Y is

$$f(x, y) = h(y \mid x) f_1(x) = \begin{cases} e^{-x} \dfrac{e^{-3x}(3x)^y}{y!} & \text{for } x > 0 \text{ and } y = 0, 1, 2, \cdots, \\ 0 & \text{elsewhere,} \end{cases}$$

$$= \begin{cases} \dfrac{e^{-4x} x^y}{y!} 3^y & \text{for } x > 0 \text{ and } y = 0, 1, 2, \cdots, \\ 0 & \text{elsewhere.} \end{cases}$$

The marginal p.f. of Y is

$$f_2(y) = \begin{cases} \displaystyle\int_0^\infty \frac{e^{-4x} x^y}{y!} 3^y \, dx & \text{for } y = 0, 1, 2, \cdots, \\ 0 & \text{elsewhere.} \end{cases}$$

Now, $\displaystyle\int_0^\infty \frac{e^{-4x} x^y}{y!} 3^y \, dx$

$$= \frac{3^y}{y!} \int_0^\infty e^{-4x} x^{y+1-1} \, dx$$

$$= \frac{3^y}{y!}\frac{\Gamma(y+1)}{4^{y+1}} = \frac{3^y}{4^{y+1}} = \frac{1}{4}\left(\frac{3}{4}\right)^y.$$

Thus, $f_2(y) = \begin{cases} \frac{1}{4}\left(\frac{3}{4}\right)^y & \text{for } y = 0, 1, 2, \cdots, \\ 0 & \text{elsewhere.} \end{cases}$

The conditional p.d.f. of X, for $x > 0$, given $Y = y$ is

$$g(x \mid y) = \frac{f(x, y)}{f_2(y)} = \frac{\frac{e^{-4x}x^y}{y!}3^y}{(1/4)(3/4)^y}.$$

Thus, $g(x \mid 0) = \dfrac{\frac{e^{-4x}x^0}{0!}3^0}{(1/4)(3/4)^0} = 4e^{-4x}$.

Hence, $g(x \mid 0) = \begin{cases} 4e^{-4x} & \text{for } x > 0 \\ 0 & \text{elsewhere.} \end{cases}$

11.14 Interpreting a Probability Distribution

Coin Tossing Assume that a balanced coin is tossed four times and X denotes the total number of heads obtained. Then X is a discrete random variable which can assume values 0, 1, 2, 3 and 4. The probability distribution of X is shown in the Table 11.8(a):

Table 11.8 Probability distribution of random variable X

Table 11.8(a)

No. of heads x	Probability $P(X = x)$
0	0.0625
1	0.2500
2	0.3750
3	0.2500
4	0.0625
	1.000

Table 11.8(b)

No. of heads x	Frequency f	Proportion f/10000
0	616	0.0616
1	2556	0.2556
2	3793	0.3793
3	2455	0.2455
4	580	0.0580
	10000	1.0000

Suppose we repeat the experiment of observing the number of heads, X, obtained in four tosses of a balanced coin a large number of times. Then the proportion of those times in which, say, no heads are obtained $\{X = 0\}$ should approximately equal to $P(X = 0)$. The same statement holds for the other four possible values of X. These facts can be verified using simulations.

We use a computer to simulate 10000 observations of the random variable X. Table 11.8(b) shows the frequencies and the proportion for the number of heads in the 10000 observations. As expected, the proportions in the third column in Table 11.8(b) are fairly close to the true probabilities in the second column in Table 11.8(a). We compare the proportion of histogram to the probability histogram of the random variable X to see the result more easily.

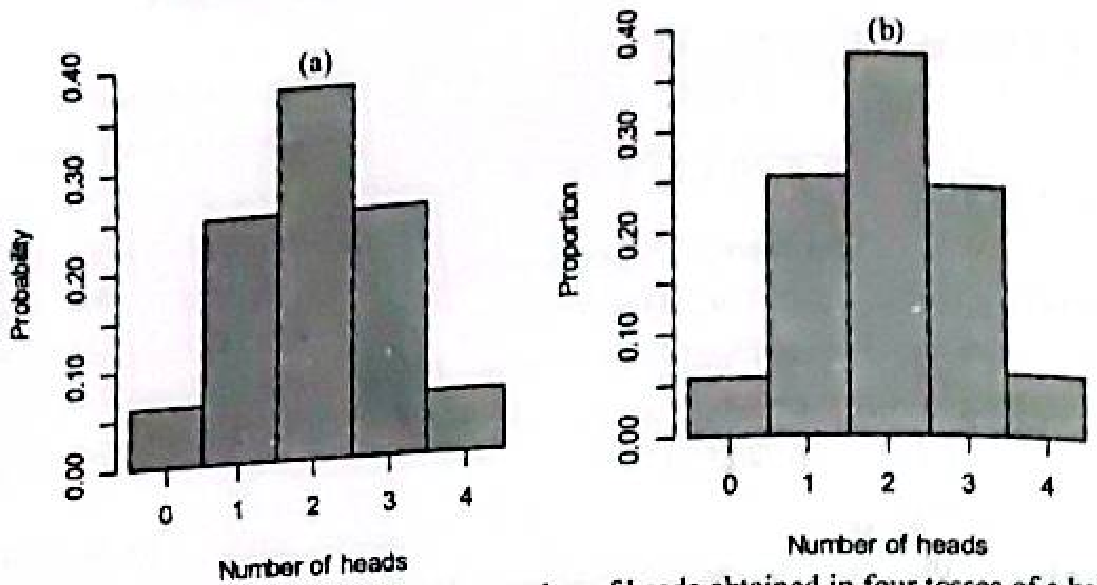


Figure 11.14.1 (a) probability histogram for the number of heads obtained in four tosses of a balanced coin; (b) Histogram of proportions for the numbers of heads obtained in four tosses of a balanced coin for 10000 observations

If, say 1,00,000 observations would be simulated instead of 10,000, the proportion that would appear in the third column of Table 11.8(b) would most likely be even closer to the true probabilities listed in the second column of Table 11.8(a).

Key Terms

Random Variable: A real-valued function defined on a sample space.

Discrete Random Variable: Can assume a sequence of separated points on a real line

Continuous Random Variable: Can assume any value in an interval on a real line

Discrete Probability Function: The probability function $f(x)$ of a discrete random variable X is defined as $f(x) = P(X = x)$ for every real number x.

Probability Density Function: A function $f(x)$ of a continuous random variable X defined on a real line such that for every real interval C,

$$P(X \in C) = \int_C f(x)dx.$$

Cumulative Distribution Function: The distribution function $F(x)$ of a random variable X is defined as a function

$$F(x) = P(X \le x), \ -\infty < x < \infty.$$

Bivariate Random Variable: A function that assigns an ordered pair of real numbers to each outcome in the sample space

Joint Probability Function: The collection of all probabilities $P[(X,Y) \in C]$ for all sets C of pairs of real numbers such that $\{(X,Y) \in C\}$ is an event

Discrete Joint Probability Function: The function $f(x,y)$ of a discrete bivariate random variable (X,Y) with spaces R_x and R_y defined by

$$f(x,y) = P(X = x, Y = y) \text{ for all } (x,y) \in R_x \times R_y.$$

Continuous Joint Probability Density Function: The function $f(x,y)$ of a continuous bivariate random variable (X,Y) defined as

$$f(x,y)=P(X=x,Y=y)$$

for all $(x,y)\in R_x \times R_y$.

Joint Probability Density Function: A function $f(x,y)$ of a discrete bivariate random variable (X,Y) defined on xy-plane such that for every subset C of the plane

$$P[(X,Y)\in C]=\iint_C f(x,y)dxdy.$$

Marginal Probability Function: Let $f(x,y)$ be a joint discrete probability function. The marginal probability function of X is

$$f_1(x)=\sum_y f(x,y) \text{ for all } x.$$

The marginal probability function of Y is

$$f_2(y)=\sum_x f(x,y) \text{ for all } y.$$

Marginal Density Function: Let $f(x,y)$ be a joint density function. The marginal density function of X is

$$f_1(x)=\int_{-\infty}^{\infty} f(x,y)dy \text{ for } -\infty < x < \infty.$$

The marginal density function of Y is

$$f_2(y)=\int_{-\infty}^{\infty} f(x,y)dx \text{ for } -\infty < y < \infty.$$

Conditional Probability Function: The conditional probability function of X, given $Y=y$ such that $f_2(y)>0$, is defined as

$$g(x\mid y)=\frac{f(x,y)}{f_2(y)}, \text{ for all } x.$$

Independent Random Variables: Two random variables are said to be independent of their joint probability function is equal to the product of their marginal probability functions.

Exercises 11

1. What is a random variable? Generate a random variable from an experiment of tossing a fair coin four times. How would you distinguish a discrete random variable from a continuous one? Give examples.
2. Provide an example of a random variable that does not arise from a quantitative variable of a finite population in the context of randomness.
3. Identify the following random variables as discrete or continuous with justification:
   a. Total number of points scored in a football game
   b. Blood pressure of a person
   c. Height of the ocean's tide at a given location
   d. The number of automobile accidents per year in Dhaka city

e. Increase in length of life attained by a cancer patient as a result of surgery
f. The number of telephone calls received in a telephone booth in a given time period
g. The return on an investment
h. The lifetime of a flashlight battery
i. The number of siblings of a randomly selected student
j. Number of broken eggs in a randomly chosen box
k. The number of bits in error in a transmitted message
l. The current in a copper wire
m. The length of a manufactured part
n. The length of time you have to wait at the bus stop
o. The height of a plant given a new fertilizer
p. The number of fleas on a prairie dog in a colony
q. Number of people in a household
r. The fuel efficiency (mpg) of an automobile
s. The number of cars crossing a bridge on a given day
t. The time spent by a physician examining a patient

4. What is probability distribution? How does a probability distribution differ from a frequency distribution? What rule of probability permits you to obtain any probability for a discrete random variable by simply knowing its probability distribution?
5. Define discrete probability function. Suppose that a box contains five red balls and three blue balls. If four balls are selected at random, without replacement, determine the p.f. of the number of red balls that will be obtained.
6. What is a density function? What are the conditions that a density function must have to satisfy? What is a distribution function? Write down the properties of a distribution function.
7. Define joint probability function, marginal probability function and conditional probability function with examples. When would you say two random variables independent?

Review Problems

1. Suppose that a random variable X has a discrete distribution with the following p.f.:

$$f(x)=\begin{cases} kx & \text{for } x=0,1,2,\cdots,6 \\ 0 & \text{elsewhere.} \end{cases}$$

   a. Determine the value of the constant k.
   b. Determine distribution function and use it to evaluate $P(1 < X < 4.5)$.

2. Suppose that two balanced dice are rolled, and let X denote the absolute value of the difference between the two numbers that appear. Determine the probability function. Determine distribution function and sketch it.

3. Suppose that the p.d.f. of a random variable X is as follows:

$$f(x)=\begin{cases} \frac{3}{2}\left(1-x^2\right) & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Sketch the p.d.f. and determine the values of the following probabilities:
   a. $P(X < 0.5)$, b. $P(0.5 < X < 0.75)$, c. $P(X > 0.75)$.

4. Suppose that X has the p.d.f.

$$f(x)=\begin{cases}3x^2 & \text{if } 0<x<1,\\ 0 & \text{elsewhere.}\end{cases}$$

a. Show that the area under the density curve is 1.

Find $P(0.5<X<2)$, and $P(0.5<X<0.75)$.

5. Suppose that the p.d.f. of a random variable X is as follows:

$$f(x)=\begin{cases}\frac{1}{6}\left(4-x^2\right) & \text{for } -3<x<3,\\ 0 & \text{elsewhere.}\end{cases}$$

a. Find distribution function of X. Using distribution function, determine the following probabilities: i) $P(X<0)$, ii) $P(-1<X<1)$.

6. Suppose that the p.d.f. of a random variable X is as follows:

$$f(x)=\begin{cases}\frac{2}{9}x & \text{for } 0<x<3,\\ 0 & \text{elsewhere.}\end{cases}$$

a. Find the value of t such that $P(X\le t)=\frac{1}{4}$.

b. Find the value of t such that $P(X\ge t)=\frac{1}{2}$.

7. Suppose that a random variable X can take only the values -2, 0, 1, and 4, and that the probabilities of these values are as follows: $P(X=-2)=0.2$, $P(X=0)=0.3$, $P(X=1)=0.1$, $P(X=4)=0.4$. Find the c.d.f. and sketch the c.d.f.

8. A box contains five slip of papers, marked Tk100, Tk100, Tk100, Tk500 and Tk1000. The winner of a contest selects two slip of papers at random and then gets the larger of the Taka amounts of the slips. Define a random variable Y that denotes the amount awarded. Determine the probability distribution of Y.

9. The p.m.f. of a random variable X is

$$f(x)=\begin{cases}\frac{x+k}{10} & \text{for } x=-2,-1,0,1,2,\\ 0 & \text{elsewhere.}\end{cases}$$

Determine the value of k. Determine $P(X\ge 0)$.

10. A random variable X is such that the probability of $X=x$ is proportional to $(x+1)(6-x)$ for $x=0,1,2,3,4,5$ (the possible values of X).

a. Find the p.m.f. of X.

b. Find the probability that X will be at least 3.

11. Suppose a fair die is tossed twice independently. Let X denote the minimum of the two numbers that appears. a. Find the p.m.f. of X, b. Find $P(X > 3)$.

12. Let X equal the number of keys that you try before you find the one that opens the door ($x = 1, 2, 3, 4, 5$). Then assign the appropriate value of X to each simple event.
a. Determine probability mass function.
b. construct a probability histogram for the p.m.f.
c. Find the probability that three or more keys are required to open the door.

13. A fair coin is tossed three times independently: let X denote the number of heads on the first toss and Y denote the total number of heads. Find the joint probability mass function of X and Y.

14. Consider the function $f(x)$ of a random variable X as follows:

$$f(x) = \begin{cases} \dfrac{x-2}{5} & \text{for } x = 1, 2, 3, 4, 5, \\ 0 & \text{elsewhere.} \end{cases}$$

Verify that whether $f(x)$ is a valid p.m.f.

15. Take a carton of six machine parts, with one defective and five nondefective. Randomly draw parts for inspection, but do not replace them after each draw, until a defective part fails inspection. Let Y denote the number of parts drawn. Find the probability mass function of Y. Find the probability that number of draws will be 3 or less.

16. A shipment of 8 similar microcomputers to a retail outlet contains 3 that are defective and 5 are non-defective. If a school makes a random purchase of 2 of these computers, find the probability distribution of the number of defectives.

17. A modem transmits a $+3$ voltage signal into a channel. The channel adds to this signal an error term that is drawn from the set $\{0, -1, -2, -3, -4\}$ with respective probabilities $\{4/12, 3/12, 2/12, 2/12, 1/12\}$
a. Determine the p.m.f. of the output X of the channel.
b. What is the probability that the output of the channel is equal to the input of the channel?
c. What is the probability that the output channel is negative?

18. The following is a probability function of a random variable X:

x	1	2	3	4	5	6	7
$f(x)$	c	$2c$	$2c$	$3c$	c^2	$2c^2$	$7c^2+c$

a. Find the value of c.
b. Evaluate $P(X \geq 5)$ and $P(X < 3)$.
c. If $P(X \leq k) > \frac{1}{2}$, What is the minimum value of k ?

19. Let $f(x)$ be the probability function of a random variable X which assumes the values x_1, x_2, x_3, x_4 such that $2P(X = x_1) = 3P(X = x_2) = P(X = x_3) = 5P(X = x_4)$; Find probability function and distribution function of X.

20. The following is the distribution function of a random variable X :

$$F(x)=\begin{cases}0 & ,x<-1,\\ \frac{1}{4} & ,-1\le x<1\\ \frac{1}{2} & ,1\le x<3\\ \frac{3}{4} & ,3\le x<5\\ 1 & ,x\ge 5\end{cases}$$

a. Find the probability function.

b. Determine $P(X\le 3)$, $P(X=3)$, $P(X\ge 1)$, $P(-0.4\le X\le 4)$, $P(X=5)$.

21. The number of newspapers that a certain hawker sells in a day follows the following probability distribution:

$$f(x)=\begin{cases}\frac{x}{c} & \text{for } x=1,\ 2,\\ \frac{5-x}{c} & \text{for } x=3,4,\\ 0 & \text{elsewhere.}\end{cases}$$

a. Find the value of c that makes $f(x)$ a probability function.

b. Find the distribution function.

c. What is the probability that the number of newspapers that will be sold tomorrow is at least 2?

22 Suppose that the error in the reaction temperature, in $^{\circ}C$, for a controlled laboratory experiment is a continuous random variable X having the following probability density function:

$$f(x)=\begin{cases}\frac{1}{3}x^2 & \text{for } -1<x<2,\\ 0 & \text{elsewhere.}\end{cases}$$

a. Verify that $f(x)$ is a density function and sketch $f(x)$.

b. Determine $P(0<X<1)$.

c. Determine distribution function of X, and using this function, determine $P(0<X<1)$.

23. Suppose that the p.d.f. of a random variable X is as follows:

$$f(x)=\begin{cases}\frac{1}{2}x & \text{if } 0<x<2,\\ 0 & \text{elsewhere.}\end{cases}$$

a. Find the distribution function of X and hence find $P(0.5<X<1)$.

b. Sketch the distribution function of X.

24. Suppose that in a certain region, the daily rainfall (in inch) is a continuous random variable X with p.d.f. $f(x)$ is given by

$$f(x)=\begin{cases}\frac{2}{9}(3x-x^2) & \text{for } 0 \le x \le 3,\\ 0 & \text{elsewhere.}\end{cases}$$

Find the probability that on a given day in that region the rainfall is
a. not more than 2 inch, **b.** more than 1.5 inch, **c.** between 1 and 2 inches, and **d.** exactly 2 inch.

25. The following is a probability density function of a continuous random variable X:

$$f(x)=\begin{cases}k(x-1) & \text{for } 2 \le x \le 6,\\ 0 & \text{elsewhere.}\end{cases}$$

a. Find the value of k; **b.** Find $P(X \le 3)$; **c.** Find $P(3 \le X \le 4)$.
d. Determine distribution function, **e.** Show that $F(4)-F(3)=P(3<X\le 4)$.

26. The following is the probability function of a random variable X:

$$f(x)=\begin{cases}\frac{c}{\sqrt{x}} & \text{for } 0<x<4,\\ 0 & \text{elsewhere.}\end{cases}$$

a. Find the value of c, **b.** Determine distribution function, and hence find $P(1<X<3)$.

27. The total lifetime (in years) of six-year-old dogs of a certain breed is a random variable whose distribution function is given by:

$$F(x)=\begin{cases}0 & \text{for } x \le 6,\\ 1-\frac{36}{x^2} & \text{for } x>6.\end{cases}$$

Find the probabilities that such a six-years old dog will live
a. beyond 10 years; **b.** less than 9 years; **c.** anywhere from 12 to 15 years.

28. The following is the distribution function of a random variable X:

$$F(x)=\begin{cases}1-(1+x)e^{-x} & \text{for } x>0,\\ 0 & \text{elsewhere.}\end{cases}$$

a. Find probability density function, $P(X \le 2)$; $P(1<X<3)$, and $P(X>4)$.

29. A random variable X has the density function

$$f(x)=\frac{c}{x^2+1} \quad \text{for} -\infty<x<\infty,$$

a. Find the value of constant c.
b. Find the probability that X^2 lies between 1/3 and 1.
c. Find the distribution function of X.

Ans. **a.** $c=1/\pi$, **b.** 1/6, **c.** $\frac{1}{2}+\frac{1}{\pi}\tan^{-1}x$.

30. The distribution function for a random variable X is

$$F(x)=\begin{cases}1-e^{-2x} & \text{for } x>0,\\ 0 & \text{for } x<0.\end{cases}$$

a. Find the density function, **b.** Find the probability that $X>2$, **c.** Find the probability that $-3\le X\le 5$.

31. Consider (X,Y) with joint p.m.f.

$$P(5,1)=P(10,1)=P(10,2)=1/10,$$
$$P(5,2)=P(10,1)=P(5,3)=2/10, \text{ and } P(10,3)=3/10.$$

Are X and Y independent?

32. Suppose that X and Y have a discrete joint distribution for which the joint p.f. is defined as follows:

$$f(x,y)=\begin{cases}\frac{1}{9}(x+y) & \text{for } x=0,1 \text{ and } y=0,1,2,\\ 0 & \text{elsewhere.}\end{cases}$$

a. Determine the marginal p.f.'s of X and Y.
b. Are X and Y independent?

33. A certain medical center has two telephone booths. For $i=0,1,2$, let p_i be the probability that exactly i telephone booths will be occupied on any Sunday evening at 7:00 P.M. Suppose that $p_i=0.2,0.5,0.3$ for $i=0,1,2$. Let X and Y denote the number of booth that will be occupied at 7:00 P.M. on two independent Sunday evenings. Determine
a. the joint p.f. of (X,Y); **b.** $P(X<Y)$; **c** $P(X=Y)$.

34. The joint probability density function of (X,Y) is given by

$$f(x,y)=\begin{cases}3(1-x-y) & \text{for } x\ge 0; y\ge 0;\ x+y<2,\\ 0 & \text{elsewhere.}\end{cases}$$

a. The marginal densities of X and Y,
b. Determine whether the two random variables are independent.

35. The joint probability density function of (X,Y) is given by

$$f(x,y)=\begin{cases}2 & \text{for } x\ge 0; y\ge 0,\ x+y<1,\\ 0 & \text{elsewhere.}\end{cases}$$

a. The marginal densities of X and the conditional density of Y given $X=x$.
b. Determine whether the two random variables are independent.

36. Consider the p.d.f. for X and Y

$$f(x,y)=\begin{cases}\frac{12}{7}\left(x^2+xy\right) & \text{for } 0\le x\le 1, 0\le y\le 1,\\ 0 & \text{elsewhere.}\end{cases}$$

a. $P(Y>X)$, **b.** marginal density of Y.

37. Suppose that the set of possible values for (X,Y) is the rectangle $D = \{(x,y): 0 \le x \le 1, 0 \le y \le 1\}$. Let the joint p.d.f. of (X,Y) is

$$f(x,y) = \begin{cases} \frac{6}{5}(x^2 + y) & \text{for } (x,y) \in D, \\ 0 & \text{elsewhere.} \end{cases}$$

a. Verify that $f(x,y)$ is a valid p.d.f.
b. Find $P(0 \le X \le 1/2, 0 \le Y \le 1/2)$.
c. Find the marginal p.d.f.s of X and Y.
d. Find $P(1/4 \le X \le 3/4)$.

38. Suppose (X,Y) takes values on the square $[0,1]\times[1,2]$ with joint p.d.f.

$$f(x,y) = \frac{8}{3}x^3 y.$$

Find the marginal p.d.f's of X and Y.

39. Suppose (X,Y) takes values on the unit square $[0,1]\times[0,1]$ with joint p.d.f.

$$f(x,y) = \frac{3}{2}(x^2 + y^2)$$

a. Find the marginal p.d.f. of X and use it to find $P(X < 0.5)$.
b. Determine the conditional p.d.f. of Y for every given value of X.
c. Determine $P(Y < 0.5 \mid X = 0.5)$.

40. The joint c.d.f. (X,Y) is

$$F(x,y) = \begin{cases} 0 & \text{for } x < 0 \text{ or } y < 0, \\ \frac{1}{2}xy(x^2 + y^2) & \text{for } 0 \le x < 1 \text{ and } 0 \le y < 1, \\ \frac{1}{2}x(x^2 + 1) & \text{for } 0 \le x < 1 \text{ and } y \ge 1, \\ \frac{1}{2}y(y^2 + 1) & \text{for } x \ge 1 \text{ and } 0 \le y < 1, \\ 1 & \text{for } x \ge 1 \text{ and } y \ge 1. \end{cases}$$

Find the marginal c.d.f. of X and use it, compute $P(X \le 0.5)$.

41. Suppose that the joint p.d.f. of two random variables X and Y is as follows:

$$f(x,y) = \begin{cases} \frac{3}{16}(4 - x - 2y) & \text{for } x > 0, y > 0 \text{ and } x + 2y < 4. \\ 0 & \text{elsewhere.} \end{cases}$$

a. Determine the conditional p.d.f. of X for every given value of Y.
b. Determine $P(1 < X < 2 \mid Y = 1)$.

42. A point X is chosen from the distribution

$$f_1(x)=\begin{cases}1 & \text{for } 0<x<1,\\ 0 & \text{elsewhere.}\end{cases}$$

After the value $X=x$ is observed, a point Y is chosen from the distribution

$$h(y\mid x)=\begin{cases}\dfrac{1}{x-1} & \text{for } x<y<1,\\ 0 & \text{elsewhere.}\end{cases}$$

a. Derive the marginal density of Y.

b. Derive the conditional distribution of X for every given value of X.

43. Assume that either of two instruments might be used for making a certain measurement. Instrument 1 yields a measurement whose p.d.f. f_1 is

$$f_1(x)=\begin{cases}3x^2 & \text{for } 0<x<1,\\ 0 & \text{elsewhere.}\end{cases}$$

Instrument 2 yields a measurement whose p.d.f. f_2 is

$$f_2(x)=\begin{cases}4x^3 & \text{for } 0<x<1,\\ 0 & \text{elsewhere.}\end{cases}$$

Suppose that one of the two instruments is chosen at random and a measurement X is made with it.

(a) Determine the marginal p.d.f. of X.

(b) If the value of the measurement is $X=1/3$, what is the probability that instrument 2 was used?

Self-Review Test

1. Suppose you decide to collect a bunch of cans of soda lime and measure the volume of soda lime in each can. Let X denote the amount of ml of soda lime in each can. What type of variable is X?
2. What type of random variable is "the number of letters in a word picked at random out of the dictionary?"
3. You decide to conduct a survey of families with three children. Let X denote the number of boys (out of 3 children) in each family. Is X a random variable, and if it is, what are all its possible values?
4. Let X denote the number of women taller than 65 inches in a random sample of 5 women. What type of variable is X? What are its possible values?
5. How can you check whether two discrete variables are independent?
6. How can you check whether two continuous variables are independent?
7. Are there random variables that are neither discrete nor continuous?
8. What are the main differences between discrete and continuous random variables?
9. What do you obtain if you take the first derivative of the c.d.f of a continuous variable?
10. How do you use the c.d.f to compute the probability of an interval?
11. If f is the density function of a random variable X, then how will you determine the CDF of X?

12. A random variable X has density function $f(x) = 1$ for $0 \le x \le 1$ and 0 otherwise.
    (i) Find $P(X \le x)$.
    (ii) Find x_a such that $P(X \le x_a) = a$.
    (iii) Sketch the p.d.f. and the c.d.f.
13. What properties does the p.d.f satisfy?
14. What properties does the p.m.f satisfy?
15. A green, a red, and a blue die are tossed. In each of the following, find the values that X can assume:
    (a) Let X denote the sum of the three dice.
    (b) Let X denote the minimum number of dots showing on any die.
    (c) Let X denote two times the number of dots on the green die.
16. For a continuous random variable X write each of the following probabilities in two different ways, one in terms of $f(x)$ and one in terms of $F(x)$:
    (a) $P(X > 5)$, (b) $P(5 < X < 10)$, (c) $P(5 < X < 10 \text{ or } X > 20)$.
17. The density function of the random variable X is given by

$$f(x) = \begin{cases} 1 - ax, & 0 < x < 2, \\ 0 & \text{otherwise.} \end{cases}$$

    (i) What is the value of a?
    (ii) Find the distribution function of the random variable X.
18. Is the following functions distribution functions?

$$F(x) = \begin{cases} 0, & x < -5, \\ x, & -5 \le x \le 0.5, \\ 1, & x \ge 0.5. \end{cases}$$

19. Let X be a continuous random variable with p.d.f.

$$f(x) = \begin{cases} k - |x|, & |x| < 0.5, \\ 0 & \text{otherwise.} \end{cases}$$

    Find the value of k.
20. Choose the correct one:
    A numerical description of the outcome of an experiment is called a
    (a) descriptive statistic, (b) probability function, (c) variance, (d) random variable.
21. Choose the correct one:
    For a continuous random variable X, the probability density function $f(x)$ represents
    (a) the probability at a given value of X
    (b) the area under the curve at X
    (c) the area under the curve to the right of X
    (d) the height of the function at X.
22. The joint p.d.f. of (X, Y) is given by

$$f(x, y) = \begin{cases} 6(1 - y), & 0 \le x \le y \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find $P(X \le 3/4, Y \le 1/2)$ and $P(X \le 3/4, Y \ge 1/2)$.